

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Prediction of In-hospital Mortality in Mechanically Ventilated Patients with Congestive Heart Failure

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ABSTRACT

In-hospital mortality among mechanically ventilated congestive heart failure CHF patients is a pressing concern, requiring accurate risk prediction to enhance clinical care. This study aimed to develop and validate a predictive model for mortality using advanced machine learning techniques and feature selection strategies.

Patient data from the MIMIC-IV database were analyzed, starting with 90 features narrowed to 60 using backward elimination. Seven machine learning models, including GATE, were evaluated through nested k-fold cross-validation. To improve performance, three GATE models trained on first, last, and mean value data were combined into an ensemble. SHAP analysis was employed to identify 20 key features, and a compact model was developed and optimized based on this streamlined feature set. Performance metrics, including AUROC, were used to evaluate model accuracy.

Data from 8,517 patient stays were analyzed. The GATE model outperformed the other six models, with the ensemble demonstrating enhanced generalizability. SHAP analysis identified critical features such as age, MV duration, SAPSII and LODS scores, lactate levels, and markers of organ dysfunction. The compact GATE model, built on 20 essential features, achieved an AUROC of 0.87, closely matching the full model's 0.88, while improving computational efficiency and interpretability.

The GATE-based prediction model provides robust and accurate identification of mortality risk in mechanically ventilated CHF patients. The compact model offers a practical balance between performance and efficiency, making it a valuable tool for clinical decision-making and resource allocation.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AUROC	Area Under the Receiver Operating Characteristic
CHF	Congestive Heart Failure
DL	Deep learning
GATE	Gated Additive Tree Ensemble
HF	Heart Failure
ICU	Intensive care unit
kNN	k-Nearest Neighbors
LODS	Logistic Organ Dysfunction System
MIMIC-IV	Medical Information Mart for Intensive Care IV
ML	Machine learning
MV	Mechanical ventilation
NODE	Neural Oblivious Decision Ensembles
OASIS	Oxford Acute Severity of Illness Score
SAPS II	Simplified Acute Physiology Score II
SHAP	SHapley Additive Explanations
SOFA	Sequential Organ Failure Assessment
XGBoost	Extreme Gradient Boosting

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Heart failure (HF) is a clinical syndrome resulting from structural or functional cardiac abnormalities, leading to symptoms such as shortness of breath, fatigue, and fluid retention. In 2021, HF was universally defined as a condition characterized by elevated levels of natriuretic peptides or evidence of pulmonary or systemic congestion [1]. Congestive heart failure (CHF), a more specific and advanced stage of HF, is characterized by significant fluid buildup and congestion. Due to its severity and complexity, CHF often requires intensive care unit (ICU) admission for management [2].

Mechanical ventilation, a critical life-support intervention, is frequently used in the ICU to manage CHF patients experiencing respiratory failure. Although life-saving, this intervention carries substantial risks, including hemodynamic instability, alveolar inflammation, and complications caused by altered intrathoracic pressures [3]. These challenges highlight the urgent need for advanced predictive tools to assess patient outcomes, aiding more effective clinical decision-making and improving management for this high-risk group.

Predictive modeling is an indispensable tool in healthcare, offering the ability to forecast patient outcomes based on clinical data, thereby enhancing decision-making. For mechanically ventilated CHF patients, predictive models assist in identifying individuals who might benefit from early interventions, intensive monitoring, or alternative therapeutic strategies. However, predicting early mortality in this population remains challenging due to the intricate interplay of systemic organ dysfunction, comorbidities, and rapidly changing clinical conditions.

Traditional scoring systems, such as SOFA, APACHE-II, LODS, MODS, and SAPS-II, are significantly associated with mortality in critically ill patients and have demonstrated utility in mortality prediction [4]. While the SOFA and GWTG-HF risk scores provide valuable insights, they have notable limitations. The SOFA score, for example, evaluates organ failure but lacks specificity for heart failure, whereas the GWTG-HF score captures HF-specific risks but insufficiently addresses systemic dysfunction. Additionally, these tools face challenges such as incomplete clinical data, including missing arterial blood gas measurements, and their static nature, which does not accommodate the dynamic changes in a patient's condition [5]. Population variability and the need for interpretable, actionable predictions

further limit their widespread applicability, underscoring the necessity for more adaptable and inclusive models.

Machine learning (ML) and deep learning (DL) have shown remarkable promise in predicting mortality among mechanically ventilated patients, including those with CHF. These computational techniques excel in analyzing complex, high-dimensional clinical data and identifying patterns that traditional scoring systems may miss. Research indicates that ML models often outperform conventional systems such as SOFA or APACHE-II, achieving higher Area Under the Curve (AUC) values for short-term outcomes like 30-day mortality [6].

This thesis aims to address the limitations of existing predictive tools by developing ML and DL models specifically tailored to predicting in-hospital mortality in mechanically ventilated CHF patients. By leveraging the rich and diverse clinical data available in the Medical Information Mart for Intensive Care IV (MIMIC-IV) database, these models will integrate multiple parameters to provide accurate, dynamic, and actionable predictions. This work seeks to improve risk stratification and guide timely clinical interventions, ultimately enhancing patient outcomes in this high-risk population.

1.2 Background and Research Problems

CHF is a clinical syndrome characterized by structural or functional impairments of the heart, leading to insufficient ventricular filling or blood ejection, as defined by the American College of Cardiology (ACC) and the American Heart Association (AHA). It is a prevalent condition worldwide, affecting approximately 26 million people, and is associated with significant morbidity and mortality. Ischemic heart disease, the leading cause of death globally, is also the primary contributor to CHF. This condition imposes a considerable burden on healthcare systems due to increased costs, reduced functional capacity, and diminished quality of life. Early diagnosis and effective treatment are essential to mitigate hospital readmissions, lower mortality rates, and enhance patient outcomes [7].

Patients with CHF are particularly vulnerable to respiratory failure, which often stems from complications such as acute pulmonary edema, hypoxemia, or hypercapnia. Acute pulmonary edema, a hallmark of decompensated CHF, occurs due to elevated left ventricular pressures causing fluid accumulation in the alveoli, resulting in severe respiratory distress [8]. In critical cases, cardiogenic shock, marked by significantly reduced cardiac output, can impair systemic oxygenation and necessitate mechanical ventilation to reduce the metabolic demands of breathing [9]. Respiratory muscle fatigue, exacerbated by the increased workload on the

respiratory system in CHF, further contributes to the need for ventilatory support [10]. Coexisting pulmonary conditions, such as infections or chronic obstructive pulmonary disease (COPD), can complicate respiratory failure, necessitating tailored patient management [11].

Mechanical ventilation (MV) plays a crucial role in managing respiratory failure in decompensated CHF patients. It serves as a life-saving intervention by taking over the breathing workload when patients can no longer sustain adequate ventilation. MV reduces ventricular preload and afterload, decreases extravascular lung water, and improves myocardial oxygenation, thereby increasing cardiac output. The positive pressure and oxygenation provided by MV also aid recovery from left ventricular dysfunction and hypoxia-induced pulmonary vascular constriction [10], [12]. Non-invasive ventilation (NIV), such as continuous positive airway pressure (CPAP), is often the first-line approach to avoid the complications of invasive MV. However, in severe cases or when NIV fails, invasive MV becomes essential to stabilize respiratory and hemodynamic parameters [13]. Despite its benefits, MV carries risks such as volutrauma, hyperoxia-induced injury, ventilator-associated infections, and weaning challenges related to persistent cardiac dysfunction [10]. Balancing these risks with the benefits is critical in optimizing outcomes for high-risk CHF patients.

Predictive modeling has become a cornerstone in healthcare for forecasting patient outcomes and guiding clinical decision-making. In the context of mechanically ventilated CHF patients, predictive models help clinicians identify those who might benefit from early intervention, intensive monitoring, or alternative therapeutic strategies. Traditional scoring systems such as SOFA, APACHE-II, LODS, MODS, and SAPS-II have been widely used to predict mortality in critically ill patients [4], [14], [15]. While these tools provide valuable insights, they have notable limitations. For example, the SOFA score evaluates organ failure but lacks specificity for CHF, whereas the GWTG-HF score focuses on CHF-specific risks but fails to account for broader systemic dysfunctions [5]. Additionally, these models rely on static clinical data, which cannot accommodate the dynamic changes observed in critically ill patients, and they often suffer from incomplete data, such as missing arterial blood gas measurements [5]. These limitations underscore the need for more adaptable and dynamic predictive models.

Recent advancements in ML and DL offer promising alternatives to traditional models. Studies have demonstrated the efficacy of algorithms such as CatBoost, Extreme Gradient Boosting (XGBoost), Decision Trees, Random Forests, Support Vector Machines (SVM), k-Nearest Neighbors (kNN) and Logistic Regression in

predicting in-hospital mortality among mechanically ventilated patients[6], [16], including CHF patients [17]–[20].

These models outperform traditional scoring systems by leveraging diverse datasets and advanced techniques. Suveen et al. [18] identified random forest (RF) as the best-performing model for HFpEF, achieving a mean C-statistic of 0.72 for mortality and 0.76 for hospitalization prediction, with key predictors including blood urea nitrogen, body mass index, and KCCQ scores. Li Le et al. [17] used the MIMIC-IV database to evaluate 12 ML algorithms for predicting prolonged mechanical ventilation (PMV) and in-hospital mortality in CHF patients, with CatBoost achieving the highest AUCs of 0.766 for PMV and 0.844 for mortality. Similarly, Hexin et al. [20] demonstrated CatBoost’s superiority among ML models, achieving an Area Under the Receiver Operating Characteristic (AUROC) of 0.863 and an F1 score of 0.747. These findings emphasize the ability of ML models to integrate a broader range of variables, offering more accurate predictions and risk stratifications than traditional scoring systems, enhancing their potential for clinical decision-making.

Despite advancements in predictive modeling for CHF patients requiring MV, several key gaps persist. Many existing models are tailored for general ICU populations, lacking the specificity needed to address the distinct clinical and pathophysiological characteristics of CHF. Furthermore, much of the existing research relies on outdated databases like MIMIC-III, which lack comprehensive historical data and fail to reflect current clinical practices, limiting the accuracy and relevance of these models. Techniques such as grid search and Bayesian optimization, while effective in enhancing performance, are computationally demanding and challenging to scale. Similarly, feature engineering, though capable of improving model accuracy by incorporating domain-specific variables, requires significant expertise and risks introducing bias if not meticulously executed. Additionally, the integration of deep learning approaches into predictive models for CHF patients remains underexplored. To address these challenges, there is an urgent need for CHF-specific, interpretable, and dynamic ML/DL models validated across diverse and up-to-date datasets. Leveraging the latest version of MIMIC-IV and employing advanced modeling techniques can enhance predictive accuracy, ultimately improving clinical outcomes for mechanically ventilated CHF patients.

1.3 Research Objectives and Conceptual Framework

1.3.1 Research Objectives

The primary objective of this study is to develop and evaluate predictive models for in-hospital mortality among (CHF) patients who require mechanical ventilation (MV) within the first 24 hours of ICU admission. To achieve this, the project employs both ML and DL approaches to compare their predictive performance and identify key clinical variables contributing to mortality. By addressing existing limitations in prior research, this study aims to provide a more robust and clinically applicable predictive framework.

The research objectives can be summarized as follows:

1. Data preparation and feature engineering:

Extract and process data from the MIMIC-IV dataset to create structured tables and new derived fields. These new variables will be analyzed to determine their impact on in-hospital mortality. Through careful feature engineering, the study will identify potential predictors among the newly created variables that may deeply influence patient outcomes.

2. Enhancing Predictive Accuracy through Advanced Models:

Build and compare various machine learning and deep learning models to predict in-hospital mortality. This comparison will determine which modeling approach provides the best predictive performance.

3. Optimize Model Performance:

Existing studies are often limited by retrospective designs and reliance on a single dataset, leading to potential overfitting and limited generalizability. To address this, the predictive models developed in this project will be validated with techniques to ensure model robustness and stability.

4. Analyze Key Variables:

For ML and DL models to be clinically useful, they must provide interpretable outputs. Identify the most significant predictors of in-hospital mortality and explore their clinical relevance to assist in early diagnosis and risk stratification.

To address the challenges identified in prior research, this project adopts the following solutions, combining both machine learning and deep learning methods while addressing each problem systematically:

1. Data collection

Beside clinical data, ICU scoring systems such as LODS (Logistic Organ Dysfunction System), SAPS II (Simplified Acute Physiology Score II), and SOFA (Sequential Organ Failure Assessment) will be included as features in the study. A limitation of previous studies is the lack of detailed comorbidity data. To address this, the study will extract comorbidity conditions for patients across 19 disease categories (e.g., stroke, diabetes, chronic obstructive pulmonary disease, acute kidney injury). By analyzing comorbidity results, the study will determine which specific conditions are significantly associated with higher in-hospital mortality rates. This analysis will enhance the models' ability to account for patient-specific risk factors, offering a more precise and clinically relevant risk assessment.

2. Feature Selection and Inclusion of Critical Clinical Variables

Identify and select the most relevant clinical and demographic features for predicting in-hospital mortality in mechanically ventilated CHF patients using advanced feature selection techniques, including backward elimination and SHapley Additive exPlanations (SHAP) analysis.

3. Develop and compare predictive models

Previous studies have shown that models such as CatBoost and Random Forest exhibit strong predictive capabilities but do not incorporate deep learning techniques. This study will expand upon existing methods by including deep learning models, such as GATE and FT-Transformer. Deep learning models are particularly effective at identifying nonlinear patterns and interactions in large datasets, making them well-suited for ICU mortality prediction tasks. By developing and systematically comparing both ML and DL models, the study will identify the best-performing method based on metrics such as AUC (Area Under the Curve).

4. Generalizability

Ensuring model generalizability is a critical aspect of this study. To achieve this, nested cross-validation will be employed to provide unbiased performance evaluation during model development. Nested cross-validation separates data into outer and inner loops, ensuring that hyperparameter tuning and performance evaluation are conducted independently.

By addressing limitations such as the absence of critical clinical and comorbidity variables, incorporating deep learning models, and ensuring model interpretability and validation, this project aims to develop a robust, generalizable, and clinically

applicable framework for predicting in-hospital mortality in CHF patients requiring mechanical ventilation. The integration of these solutions will not only improve predictive accuracy but also provide valuable insights for clinicians, ultimately enhancing patient care and outcomes.

1.3.2 Conceptual Framework

The conceptual framework of this study integrates several key components to achieve the research objectives:

1. Dataset and Feature Selection

Data from the MIMIC-IV database were used, encompassing a wide range of clinical and demographic variables. Feature selection methods, including backward elimination and SHAP analysis, were employed to filter 90 initial variables down to 60 and, subsequently, to 20 key features critical for model development.

2. Model Development and Evaluation

Seven machine learning and deep learning models, including Logistic Regression, XGBoost, CatBoost, and GATE, were developed and evaluated. Nested k-fold cross-validation was used to ensure robust training and validation, and an ensemble model approach was explored to combine predictions from different data aggregation strategies.

3. Compact Model Design

A compact version of the best-performing GATE model was designed using 20 key features identified by SHAP. This model aims to balance predictive performance with computational efficiency and clinical interpretability.

4. Validation and Comparison

Both the full and compact models were validated and compared using metrics such as AUROC. This step ensures that the compact model retains sufficient predictive power while being more practical for clinical use.

5. Clinical Relevance and Usability

The framework emphasizes the clinical relevance of the identified features, ensuring that the model aligns with real-world workflows. Insights into the importance of features such as MV duration, age, SAPSII score, and organ dysfunction markers were highlighted to guide clinical decision-making.

This conceptual framework ensures a systematic approach to developing and

validating a robust predictive model, integrating both technical rigor and clinical applicability to address a critical healthcare challenge.

1.4 Contributions

The contributions of this project are as follows:

1. The project proposes a data preprocessing pipeline to collect data from MIMIC-IV dataset, clean and prepare the data, including noise and outlier removal, missing value imputation.
2. New variables, such as ICU scoring systems (SOFA, LODS, SAPS II), are integrated to enhance model performance. Additionally, the study evaluates the impact of 19 comorbidity conditions on in-hospital mortality to identify high-risk patient groups.
3. The model was rigorously validated using nested k-fold cross-validation and evaluated against multiple metrics, ensuring robust performance and reliability. The study also demonstrated the trade-offs between a full-feature model and a compact model, providing insights into model optimization.
4. Key feature importance analysis is conducted using techniques such as SHAP, enabling the identification of critical predictors that influence mortality outcomes.
5. By focusing on clinically relevant features, this thesis not only highlights key risk factors for mortality in CHF patients but also provides a model that aligns with real-world clinical workflows. This enhances the model's potential for practical adoption in healthcare settings.

1.5 Organization of Thesis

Chapter 2 provides an overview of the (i) scope of this work, (ii) foundational background on both the clinical and computational aspects, and (iii) prior literature closely related to this project. Chapter 3 outlines the methodology employed in this research, including a schematic overview of the overall approach and key components such as data extraction and machine learning model training. Chapter 4 presents the experimental results and compares the performance of different model developments. Chapter 5 offers a discussion of the observations and insights from this work, integrating both clinical and computational perspectives. Finally, Chapter 6 offers concluding remarks and outlines directions for future research.

CHAPTER 2. LITERATURE REVIEW

In this section, we present the scope of research in this work and how it is situated within the broader context of existing literature. We then provide background and related foundational knowledge in both computer science and biomedical domains. Finally, we highlight prior studies closely related to this work, which have partly inspired the motivation for our research.

2.1 Scope of Research

This work focuses on predicting in-hospital mortality for patients with CHF who required MV within the first 24 hours of admission to the ICU. The study utilizes data extracted from the MIMIC-IV database (version 3.0), a comprehensive, de-identified dataset of ICU patients, covering a large population for over 20 years, from June 2001 to October 2022.

The scope of this research includes the following key aspects:

Study Population. The study population for this project was defined based on specific inclusion criteria to ensure the selection of a relevant and focused cohort. Participants included in the analysis met the following criteria:

- adults aged over 18 years
- with CHF identified as the primary cause of admission to the ICU
- had to have received MV within the first 24 hours of their ICU admission

Data. The study leverages structured data (e.g., demographics, vital signs, laboratory test results, comorbidities, ICU scores like SOFA, SAPS II, and LODS) from the MIMIC-IV database.

Models. The research develops and compares traditional machine learning models (e.g., Random Forest, CatBoost) and advanced deep learning models (e.g., GATE, FT-Transformer).

Expected Outcomes. There are two expected outcome from this work as follows:

- The primary outcome is the prediction of in-hospital mortality.
- The study identifies and analyzes key clinical variables contributing to mortality predictions, providing insights into patient risk factors.

Clinical Impacts.

- The models aim to provide early identification of high-risk CHF patients in

ICU settings, assisting clinicians in making timely treatment decisions.

- The research also supports efficient resource allocation by prioritizing care for patients with higher mortality risk.

By defining a clear and focused population, utilizing a large and well-documented dataset, and applying both machine learning and deep learning methods, this research provides a comprehensive and clinically relevant approach to improving mortality prediction for critically ill CHF patients.

2.2 Background

2.2.1 Foundation Knowledge of Clinical Background

This section outlines the foundational background on CHF, mechanical ventilation, and in-hospital mortality. As this thesis focuses on the intersection of these critical areas, the following discussion aims to provide a comprehensive understanding of the clinical context and the motivation driving this project.

a. Overview of Congestive Heart Failure

HF is an increasingly prevalent public health issue and now a global epidemiology. HF affects an estimated 64 million individuals worldwide, according to a 2023 health article [21] from the Cardiac Failure Review (CFR). HF remains a major health challenge, marked by high mortality and morbidity rates, poor quality of life, and reduced functional capacity, while placing a heavy demand on healthcare system. The results by Shahim et al. [21] stated that HF's prevalence, incidence, mortality, and morbidity rates vary geographically, impacted by the different aetiologies and clinical profiles observed among patients with HF.

HF is defined by Bozkurt et al. [1] in the Journal of Cardiac Failure as follows: “HF is a clinical syndrome with symptoms and/or signs caused by a structural and/or functional cardiac abnormality and corroborated by elevated natriuretic peptide levels and/or objective evidence of pulmonary or systemic congestion.” This work categorizes HF into four stages. At Risk for HF (Stage A) refers to patients who are at risk for heart failure but have no current or prior symptoms or signs of HF and no structural or biomarker evidence of heart disease. Pre-Heart Failure (Stage B) applies to patients without current or prior symptoms or signs of HF but with evidence of structural heart disease, abnormal cardiac function, or elevated natriuretic peptide levels. Heart Failure (Stage C) describes patients with current or prior symptoms and/or signs of HF caused by structural and/or functional cardiac abnormalities. Advanced Heart Failure (Stage D) includes patients with severe symptoms and/or signs of HF at rest, frequent hospitalizations despite guideline-

directed medical therapy (GDMT), refractory or intolerant to GDMT, and requiring advanced therapies such as heart transplant, mechanical circulatory support, or palliative care.

In my thesis work, I focus on Congestive Heart Failure, or CHF. The term "congestive" specifically refers to the buildup of fluid (congestion) in the lungs and other body tissues, which is a common symptom in many HF patients. However, not all HF cases involve congestion. Formally, CHF, as defined by the American College of Cardiology (ACC) and the American Heart Association (AHA), is “a complex clinical syndrome that results from any structural or functional impairment of ventricular filling or ejection of blood.” [7]. CHF can arise from any disorder that impairs ventricular filling or ejection of blood to the systemic circulation. Patients typically present with symptoms such as fatigue, dyspnea, reduced exercise tolerance, and signs of systemic or pulmonary congestion.

b. Mechanical Ventilation and In-Hospital Mortality

Among the numerous short-term life support techniques, MV stands out as the most widely used worldwide. MV technique is applied daily for a broad range of indications, ranging from surgical procedures to acute organ failure [3]. While it is a lifesaving technique, it also introduces important potential complications, including ventilation-induced lung injury among other worse outcomes from patient-ventilator dyssynchronies. Pham et al. [3] stated that minimizing the duration of MV is the most effective way to reduce complications. As soon as ventilation begins, clinicians should evaluate how and when to discontinue its use, while continuously determining the most appropriate weaning strategy.

For patients with HF, MV is a life-saving intervention, especially in cases with acute decompensated HF. MV can help decreasing ventricular preload and afterload, reducing extra-vascular lung water, and easing the work of breathing in HF [10]. However, effective management requires a thorough understanding of both the benefits and risks of invasive MV to maximize its advantages while minimizing adverse effects, such as volutrauma, hyperoxia-induced injury, and difficulty assessing readiness for liberation. When applied carefully, both invasive MV and non-invasive positive pressure ventilation are important tools in managing respiratory failure in patients with acute decompensated HF [10]. In a similar observation, Magliocca and Ristagno [22] show that the use of continuous positive airway pressure (CPAP) and non-invasive ventilation (NIV) has consistently been shown to reduce in-hospital mortality and endotracheal intubation rates. It also improves respiratory distress by lowering respiratory rate and increasing oxygenation

without significantly affecting hemodynamics, compared to standard medical therapy.

In-hospital mortality refers to the rate of death among patients during their stay in a hospital [23]. Though it can serve as a valuable indicator of care quality, but careful statistical analysis is essential to prevent mistakenly linking mortality variations to differences in health care when they are actually driven by case mix disparities. In a research work with inpatient surgery [24], Ghaferi et al. demonstrated that variations in overall mortality for individual operations were primarily driven by differences in death rates among patients with major complications.

2.2.2 Machine Learning and Predictive Models

a. Artificial intelligence

Artificial Intelligence (AI) refers to a branch of technology that enables machines and computers to contain or entail (human-like) intelligence, such as learning, understanding, problem-solving, decision-making, creativity, and autonomous behavior [25], [26].

AI-powered applications and devices are capable of perceiving and recognizing objects, interpreting and responding to human language, and learning from data and experiences. These systems can provide detailed recommendations to users or experts and act independently to replace or minimize the need for human intervention [27]. A notable example of such autonomy is the self-driving car, which uses AI to navigate and make real-time decisions without human control.

In 2024, however, the majority of AI advancements, research efforts, and public attention are centered around generative AI (gen AI). Generative AI represents a powerful technology that can generate original content, such as text, images, videos, and other creative outputs [28]. To fully grasp the significance of generative AI, it is essential to first understand the foundational technologies it relies on—machine learning and deep learning.

These underlying technologies serve as the backbone of generative AI tools, enabling them to learn patterns, process massive amounts of data, and produce novel content that closely resembles human-generated results.

b. Machine learning

Machine Learning is a subset of artificial intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without explicit programming. Instead of being given detailed rules, ML systems analyze data,

identify trends, and improve performance over time through experience [29]. Machine learning can be broadly categorized into three main types [30]:

Supervised Learning

In supervised learning, the model is trained on labeled data, where both input features and the corresponding output labels are provided. The goal is to learn the mapping function between inputs and outputs for tasks like classification (e.g., predicting whether an email is spam) and regression (e.g., predicting house prices based on features like area and location). Example algorithms include: Linear Regression, Logistic Regression, Support Vector Machines (SVMs), and Random Forests.

Unsupervised Learning

In unsupervised learning, the model works with unlabeled data to identify hidden patterns, relationships, or structures. It is commonly used for clustering (e.g., customer segmentation) and dimensionality reduction (e.g., reducing features for visualization). Example algorithms include: K-Means Clustering, Principal Component Analysis (PCA), and Autoencoders.

Reinforcement Learning

Reinforcement learning focuses on training models to take actions in an environment to maximize cumulative rewards. The model learns through trial and error, receiving rewards or penalties based on its actions. This approach is widely used in robotics, game playing, and self-driving cars. Example techniques include: Q-Learning, Deep Q-Networks (DQN), and Policy Gradient Methods.

c. Deep learning

Deep Learning is a specialized subset of machine learning that leverages artificial neural networks (ANNs) to process data and extract complex features [29]. Inspired by the structure and function of the human brain, deep learning uses multiple layers of interconnected neurons to automatically learn representations from raw data. This makes deep learning particularly powerful for tasks involving large, high-dimensional datasets.

The followings are some key concepts in deep learning

Artificial Neural Networks (ANNs)

An ANN consists of an input layer, one or more hidden layers, and an output layer. Each layer contains nodes (neurons) that process and transform data using

activation functions like ReLU, Sigmoid, or Softmax.

Training and Backpropagation

Neural networks learn by minimizing a loss function using optimization techniques like Gradient Descent. During training, the backpropagation algorithm calculates gradients to adjust the weights of connections between neurons, reducing prediction errors.

Common Deep Learning Architectures:

- **Convolutional Neural Networks (CNNs)** [31]: CNNs are designed for processing images and spatial data. They use convolutional layers to automatically extract features like edges, shapes, and textures. CNNs are widely used in image recognition and object detection tasks.
- **Recurrent Neural Networks (RNNs)** [32]: RNNs are well-suited for sequential data, such as time series and natural language processing (NLP). By incorporating feedback loops, RNNs can maintain memory of previous inputs. Advanced variants like LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit) overcome challenges like vanishing gradients, making them effective for tasks like text generation and speech recognition.
- **Transformers**: [33]: Transformers have revolutionized NLP by introducing the self-attention mechanism, which enables parallel processing of sequential data. Models like BERT and GPT are based on transformer architecture and excel at tasks like machine translation, text summarization, and sentiment analysis.
- **Generative Adversarial Networks (GANs)** [34]: GANs consist of two networks: a generator and a discriminator. The generator creates synthetic data, while the discriminator evaluates its authenticity. GANs are widely used for image generation, data augmentation, and video synthesis.

d. General stages in a ML pipeline

This section outlines the general stages conducted in this project, and subsequent sections will detail the specific techniques and implementations. Developing a ML or DL model involves a series of systematic steps to ensure effective learning, generalization, and performance on real-world data. The following steps are collected and extracted from the work of Goodfellow et al. [35], and Janiesch and Zschech [30].

1. **Data Collection** Data is the foundation of any ML or DL project. In this step:

- Gather data from relevant sources such as databases, sensors, APIs, or public datasets
 - For structured data: Collect rows and columns representing features and target labels.
 - For unstructured data: Gather images, text, videos, or audio files.
2. **Data Preprocessing** Raw data often contains inconsistencies or missing values and may not be suitable for training directly. Preprocessing ensures that data is clean, consistent, and ready for modeling:
- **Handling Missing Data:** Use techniques such as mean/mode imputation, interpolation, or advanced methods like k-NN imputation.
 - **Feature Scaling:** Standardize or normalize data using methods like Min-Max Scaling or Z-score Scaling to ensure features are on a similar scale.
 - **Encoding Categorical Data:** Convert categorical variables into numerical values using techniques such as One-Hot Encoding or Label Encoding.
 - **Outlier Detection and Removal:** Detect and address outliers using statistical techniques or domain knowledge.
 - **Feature Engineering:** Create new meaningful features from raw data and apply domain knowledge to transform or combine existing features.
3. **Exploratory Data Analysis (EDA)** EDA is the process of analyzing and visualizing the data to identify patterns, relationships, and distributions. This step includes:
- **Statistical Analysis:** Compute measures like mean, median, variance, and correlation.
 - **Data Visualization:** Use tools such as histograms, boxplots, scatter plots, and heatmaps to explore relationships.
 - **Feature Selection:** Identifying and retaining the most relevant features is a critical step in building effective machine learning models. Techniques like Chi-Square Test and Recursive Feature Elimination (RFE) are commonly used for feature selection
4. **Model Training** During training, the model learns patterns from the training dataset by minimizing the loss function:
- (a) **Loss Function:** A mathematical function that measures prediction error (e.g., Cross-Entropy for classification, Mean Squared Error for regression).

- (b) **Optimization:** Use optimization algorithms like Stochastic Gradient Descent (SGD), Adam, or RMSProp to update model parameters.
- (c) **Regularization:** Techniques like L2 Regularization, Dropout, and Early Stopping are applied to prevent overfitting.
- (d) **Hyperparameter Tuning:** Hyperparameters are model-specific settings (e.g., learning rate, tree depth, batch size) that are not learned from the data. Use methods to search for parameter combinations.

For Deep Learning, backpropagation is used to compute gradients, adjust weights, and minimize the loss function.

5. **Model Evaluation** Evaluate the model's performance using a test set and appropriate metrics [36]:

- **Accuracy**

Accuracy is the most basic and commonly used metric. It measures the ratio of correctly predicted samples to the total number of samples:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (2.1)$$

- **Confusion Matrix [37]**

The confusion matrix is a table used to describe the performance of a classification model on a set of test data. It breaks down predictions into four categories:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 2.1: Confusion Matrix

- **Precision**

Precision measures the proportion of positive predictions that are actually correct. It is also referred to as Positive Predictive Value:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2.2)$$

Precision is used in practical medical diagnosis to minimize the number of healthy patients incorrectly classified as having a disease.

- **Recall (Sensitivity or True Positive Rate)**

Recall measures the proportion of actual positives that are correctly identified by the model:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.3)$$

Recall is used in practical medical diagnosis to ensure that fewer patients with the disease are incorrectly classified as healthy.

- **F1-Score**

The F1-Score is the harmonic mean of precision and recall, providing a balanced measure of the two:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.4)$$

Recall is used in practical medical diagnosis to ensure that fewer patients with the disease are incorrectly classified as healthy.

- **Specificity (True Negative Rate)**

Specificity measures the proportion of actual negatives that are correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (2.5)$$

- **ROC-AUC (Receiver Operating Characteristic - Area Under Curve)**

The ROC-AUC score evaluates the ability of a model to distinguish between classes by plotting the True Positive Rate (Recall) against the False Positive Rate (FPR) at different thresholds.

- **ROC Curve:** The curve shows the trade-off between recall (sensitivity) and specificity as the decision threshold changes.
- **AUC (Area Under Curve):** The ROC AUC score ranges from 0 to 1, where 0.5 indicates random guessing, while a score close to 1 represents excellent discrimination between classes.

e. Loss function

A loss function measures the difference between the model's predictions and the actual target values [38]. It is an essential component in the training process as it guides the optimization of model parameters to minimize errors. Common loss

functions include

- **Mean Squared Error (MSE)** [38]: Used in regression tasks to minimize the squared difference between predicted and actual values.

$$L = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- **Cross-Entropy Loss** [35]: Cross-Entropy Loss (also known as Log Loss) is a widely used loss function for classification problems, particularly for binary and multi-class classification tasks. It measures the difference between the predicted probability distribution and the true distribution of class labels, quantifying how well the model's predictions match the actual labels.

For Binary Classification:

In binary classification, where the target label y can be 0 or 1, the Cross-Entropy Loss is calculated as:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$$

Where:

- N : Total number of samples.
- y_i : Actual label for sample i (1 for positive, 0 for negative).
- $\hat{y}_i$: Predicted probability that the label is 1.
- $\log$: Natural logarithm function.

For Multi-Class Classification:

In multi-class classification, where C is the total number of classes, the loss function becomes:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \cdot \log(\hat{y}_{i,c})$$

Where:

- $y_{i,c}$: True class label for class c (1 if true, 0 otherwise).
- $\hat{y}_{i,c}$: Predicted probability for class c for the i -th sample.

Cross-Entropy Loss works by comparing the predicted probability distribution of a model to the actual class labels, penalizing predictions that deviate significantly

from the true values. For binary classification, where the target labels are either 0 or 1, the loss is calculated based on the predicted probability of the positive class. If the model predicts a probability close to 1 for a true positive label, the loss will be small, indicating a confident and correct prediction. Conversely, if the predicted probability is far from the true label, such as predicting 0.1 when the actual label is 1, the loss will be large, signaling poor performance. For multi-class classification, the loss extends to compare the probabilities of all classes using a softmax output, focusing on the predicted probability for the true class. By minimizing the cross-entropy loss during training, the model is encouraged to produce probabilities that are closer to 1 for the correct class while reducing the probabilities of incorrect classes. This makes Cross-Entropy Loss particularly effective for classification tasks as it provides a smooth and differentiable gradient that optimization algorithms, such as gradient descent, can use to improve model predictions iteratively.

f. Scaling (Normalization/Standardization)

Scaling refers to the process of transforming features to ensure they are on the same scale, which improves the performance of machine learning algorithms that are sensitive to feature magnitudes (e.g., Logistic Regression, KNN, Neural Networks) [39], [40]. Common methods include:

- **Min-Max Scaling:** Rescales features to a range [0, 1].

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

- **Standardization (Z-score scaling):** Centers the data to have a mean of 0 and a standard deviation of 1.

$$X' = \frac{X - \mu}{\sigma}$$

Scaling ensures that no single feature dominates the learning process due to differences in magnitude.

g. Chi-Square Test

The Chi-Square Test is a statistical method used to determine whether there is a significant association between categorical variables. It measures the discrepancy between observed and expected frequencies in a contingency table and evaluates whether the differences are due to chance or a true relationship [41], [42]. This test is widely applied in machine learning for feature selection, where it identifies the relationship between input features and target variables, particularly in classification tasks.

How the Chi-Square Test Works

1. Hypotheses:

- **Null Hypothesis** (H_0): The two variables are independent (no relationship exists).
- **Alternative Hypothesis** (H_1): The two variables are dependent (a relationship exists).

2. **Statistic Formula:** The Chi-Square test statistic (χ^2) is calculated as:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

- O_i : Observed frequency for category i .
- E_i : Expected frequency for category i .

Steps to Perform the Chi-Square Test

The following presents the sequential steps to perform the Chi-Square test

1. Create a Contingency Table

A table that displays the frequency distribution of the variables.

2. Calculate Expected Frequencies

The expected frequency for each cell in the table is calculated as:

$$E_i = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

3. Compute the Chi-Square Statistic

Use the observed and expected frequencies in the formula to calculate the Chi-Square statistic:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

- O_i : Observed frequency for category i ,
- E_i : Expected frequency for category i .

4. Determine the Degrees of Freedom (df)

For a contingency table, the degrees of freedom are:

$$df = (r - 1) \cdot (c - 1)$$

where:

- r : Number of rows,
- c : Number of columns.

5. Compare with Critical Value or p-value

The Chi-Square statistic is compared to a critical value from the Chi-Square distribution table for the given degrees of freedom and significance level (α , often 0.05). Alternatively, a p-value is computed:

- If $p < \alpha$, the null hypothesis is rejected (variables are dependent).
- If $p \geq \alpha$, the null hypothesis is accepted (variables are independent).

h. Cross-Validation

Cross-validation (CV) is a resampling technique used to evaluate the performance of machine learning models [35]. It divides the dataset into subsets to ensure that the model is assessed on unseen data, reducing the risk of overfitting. Cross-validation provides a reliable estimate of a model's generalization ability.

Working of Cross-Validation

1. The dataset is split into k equal-sized subsets (folds).
2. For each iteration:
 - One fold is used as the validation set (testing data).
 - The remaining $k - 1$ folds are used for training the model.
3. This process repeats k times, and each fold serves as the validation set exactly once.
4. The final model performance is calculated by averaging the results (e.g., accuracy, precision) across all k iterations.

Mathematical Representation

Let D be the dataset, D_{train} the training data, and D_{val} the validation data. For k -fold cross-validation:

$$\text{CV Score} = \frac{1}{k} \sum_{i=1}^k \text{Performance}(\text{Model}_i, D_{val,i})$$

Where:

- k : Number of folds.
- $D_{val,i}$: Validation subset in the i -th fold.
- Model_i : Model trained on $D_{train,i}$ (the remaining folds).

Nested Cross-Validation

Nested cross-validation (NCV) is an extension of cross-validation used for model selection and hyperparameter tuning while ensuring unbiased performance evaluation. It involves two layers of cross-validation:

- **Inner loop**: Finds the best hyperparameters for the model.
- **Outer loop**: Evaluates the model's performance using those hyperparameters on unseen data.

Working on Nested Cross-Validation

1. Split the dataset into k outer folds
2. For each outer fold:
 - Use the current outer fold as the test set, use the remaining data as the training set.
 - Perform hyperparameter tuning on the training set using an inner k -fold cross-validation
 - Split the training set into inner folds.
 - Train the model on the inner training folds and validate it on the inner validation folds.
 - Evaluate hyperparameter configurations and select the best one.
 - Train the model with the selected hyperparameters on the entire outer training set.
 - Evaluate the model on the outer test set.

3. Repeat for all outer folds.

The final performance is the average of the metrics from the outer test sets.

i. SHAP (SHapley Additive exPlanations)

SHAP, which stands for SHapley Additive exPlanations, is a widely used method for interpreting machine learning model predictions and was first introduced by Lundberg and Lee [43]. It is grounded in cooperative game theory and leverages the concept of Shapley values to fairly distribute a model's prediction to its input features. By providing feature importance values that explain how each feature contributes to the model's output, SHAP ensures transparency and interpretability in both complex and simple models. This makes it particularly useful in domains where understanding model decisions is critical, such as healthcare, finance, and risk assessment.

At its core, SHAP calculates the impact of including or excluding each feature in a prediction by evaluating all possible feature subsets. The algorithm considers every combination of input features to assess how their inclusion changes the model's prediction. Mathematically, the SHAP value for a feature i is calculated as the weighted average of its marginal contributions across all possible feature subsets. This is expressed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Here, ϕ_i is the SHAP value for feature i , F is the set of all features, S is a subset of features excluding i , and $f(S)$ represents the model's output when trained with subset S . This formula ensures a fair and consistent attribution of contributions, aligning with the principles of local accuracy, consistency, and missingness. Local accuracy ensures that the sum of SHAP values for all features equals the model's output. Consistency guarantees that if a feature's contribution increases in a model, its SHAP value will also increase. Missingness ensures that unused features are assigned a SHAP value of zero.

The meaning of SHAP lies in its ability to provide explanations for black-box machine learning models. Unlike traditional feature importance methods, SHAP not only quantifies the contribution of each feature but also explains the direction of its impact (positive or negative) on the prediction. This transparency allows practitioners to interpret model decisions effectively, identify biases, and validate the reliability of predictions. For example, in a medical diagnosis model, SHAP can

reveal which features (e.g., age, blood pressure, cholesterol level) contribute most to a diagnosis, helping clinicians understand and trust the model's decision.

In practice, SHAP values are visualized using SHAP summary plots, which display the magnitude and direction of feature contributions across all predictions. SHAP can also generate force plots and dependence plots to explain individual predictions and feature interactions. This flexibility makes SHAP a versatile tool for both global model interpretation and local instance-based explanations.

Overall, SHAP plays a crucial role in improving the interpretability and trustworthiness of machine learning models. By assigning feature attributions in a fair and transparent manner, SHAP bridges the gap between complex models and human understanding, enabling informed decision-making and fostering model accountability.

j. Predictive Models

Logistic Regression

Logistic regression [35] is a supervised machine learning algorithm designed for classification tasks, where the goal is to predict the probability that an instance belongs to a specific class or not. It is particularly suited for binary classification problems, where the output can take one of two possible classes. Logistic regression is grounded in statistical principles and analyzes the relationship between independent variables (features) and a binary dependent variable (target).

At its core, logistic regression employs the sigmoid function (also called the logistic function) to map the linear combination of input features to a probability value between 0 and 1. The sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Here, z represents the linear equation of the input features:

$$z = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

- w_0 : Bias term (intercept),
- $w_1, w_2, \dots, w_n$: Weights (coefficients) associated with each feature,
- $x_1, x_2, \dots, x_n$: Input features.

The sigmoid function maps z to a probability, where:

- If the probability is greater than 0.5 (the threshold), the instance is classified as Class 1.
- If the probability is less than or equal to 0.5, the instance is classified as Class 0.

Logistic regression is termed regression because it uses a linear regression approach to model the relationship between the features and the target variable. However, the output of the model is used for classification purposes by applying the logistic (sigmoid) transformation.

For example, consider a classification task where the goal is to determine whether a patient has a particular disease (Class 1) or not (Class 0). Given the patient's age, blood pressure, and cholesterol level as inputs, logistic regression calculates the probability of the patient belonging to Class 1. If the calculated probability exceeds 0.5, the model predicts that the patient has the disease (Class 1); otherwise, it predicts that the patient does not (Class 0).

Logistic regression is simple, computationally efficient, and interpretable, making it widely used for binary classification problems.

CatBoost

CatBoost [44], short for "Categorical Boosting," is a gradient boosting toolkit that implements Gradient Boosted Decision Trees (GBDT). It has two critical algorithmic advances: the implementation of ordered boosting and innovative algorithm for processing categorical features. It is specifically designed to handle categorical features efficiently and avoid prediction biases that commonly affect other GBDT implementations like XGBoost and LightGBM. CatBoost introduced (i) ordered boosting, a modification of standard gradient boosting algorithm, which helps to avoid target leakage, and (ii) a new algorithm for handling high cardinality categorical features. CatBoost also uses Oblivious Decision Trees (ODTs), which have a fixed structure and the same splitting criteria across levels. This structure helps reduce overfitting and speeds up both training and inference processes.

In an interdisciplinary review, Hancock and Khoshgoftaar [45] show CatBoost often outperforms other GBDT implementations, especially on heterogeneous datasets with mixed feature types. However, they also revealed that CatBoost has a sensitivity to hyper-parameter settings. Users can adjust the settings for the maximum number of iterations, the depth of decision trees, and the number of categorical feature combinations in CatBoost to balance computational resources and model performance.

FT-Transformer

FT-Transformer (Feature Tokenizer + Transformer) was first introduced by Gorishniy et al. [46] as a simple adaptation of the Transformer architecture [33] for tabular data. The model transform all features, both categorical and numerical, into embeddings and processes them through a stack of Transformer layers. As a result, each Transformer layer operates on the feature level of one object.

Gated Additive Tree Ensemble (GATE)

Gated Additive Tree Ensemble (GATE) was first introduced by Manu and Harsh [47] as a deep learning architecture for tabular data that warrants high-performance, parameter and computationally efficient. This architecture has notably (i) Gated Feature Learning Units (GFLU) for feature selection and learning, (ii) Differentiable Non-Linear Decision Trees (DNLT) as the main inductive bias, (iii) self Attention between the outputs of the trees, and (iv) additive ensemble of tree outputs. These key features facilitate representation learning while capturing the inductive bias inherent to decision trees, ensuring parameter efficiency and scalability to large datasets with thousands of features.

2.3 Related Work

In recent years, numerous studies have focused on analyzing the hospitalization process and ICU admissions of patients to uncover relationships between diseases and the use of mechanical ventilation. These studies have explored factors influencing disease progression, identified key predictors of patient outcomes, and proposed methods for prognosis to develop appropriate treatment strategies tailored to each patient's condition.

Research efforts have particularly emphasized the importance of predicting in-hospital mortality for critically ill patients, including those with CHF. By leveraging advanced ML and DL techniques, studies have demonstrated improvements in identifying risk factors and enhancing predictive accuracy.

2.3.1 Relationship Between Long-term Survival and Prolonged Mechanical Ventilation

The relationship between long-term survival and prolonged MV in critically ill patients has been an emerging focus in clinical research due to its significant public health impact [48]–[51]. Prolonged dependence on MV, typically defined as mechanical ventilation lasting more than 14 days, is associated with poor long-term outcomes, including high mortality and limited functional recovery.

A systematic review and meta-analysis by Damuth et al. [52] analyzed data from 124 studies across 16 countries, including critically ill patients who required prolonged MV, had tracheostomies for acute respiratory failure, or were admitted to ventilator weaning units. The study revealed concerning long-term survival outcomes. Specifically, the 1-year mortality rate was reported as 59% (95% CI 56–62%), increasing to 62% (95% CI 57–67%) among high-quality studies. Furthermore, in-hospital mortality was 29% (95% CI 26–32%), and only 50% of patients were successfully liberated from mechanical ventilation.

Importantly, the analysis identified significant disparities in outcomes between the USA and other international settings, particularly in post-acute care hospitals. For instance, 1-year mortality was significantly higher in the USA (73%) compared to non-USA countries (47%), while rates of successful liberation from MV were lower (47% vs. 63%). Nelson et al. [53] and Scheunemann et al. [54] also highlighted that communication about prognosis and life-sustaining treatments between physicians and critically ill patients or their families in the USA is often suboptimal. These findings emphasize the variability in care practices, resource availability, and patient management strategies across different regions.

The study in [52] also highlighted a near-linear relationship between the duration of mechanical ventilation and adverse outcomes. Prolonged MV was identified as an independent risk factor for mortality, suggesting that longer ventilation times directly contribute to increased patient mortality, poorer recovery rates, and reduced likelihood of discharge to home.

These findings underscore the need for improved strategies to identify optimal patient selection for prolonged mechanical ventilation and better integration of long-term survival data into clinical decision-making processes. Understanding the predictors and outcomes of prolonged MV is critical for developing models that support early intervention, resource allocation, and improved care pathways for patients at high risk of prolonged ventilation dependency.

This study serves as a foundation for my project, which focuses on predicting in-hospital mortality in CHF patients with early MV. By incorporating variables related to ventilation duration and other key clinical predictors, this research aims to build robust predictive models to assist clinicians in decision-making and improve patient outcomes.

2.3.2 Applying ML Algorithms in Medicine to Predict MV Outcomes

Extubation failure (EF), defined as the need for non-invasive ventilation (NIV), reintubation, or death within 48 hours following planned extubation, poses

significant risks in intensive care units (ICUs) [55]. EF can lead to increased mortality, prolonged hospital stays, and complications such as ventilator-associated pneumonia. To address these challenges, ML models have emerged as powerful tools for predicting EF, outperforming traditional clinical judgment and scoring systems [56]–[58].

A recent study by Zhao et al. [59] developed and validated a machine-learning model using the MIMIC-IV database to predict extubation failure. The study included 16,189 ICU patients and employed the CatBoost algorithm, a gradient boosting model, trained on 89 clinical and laboratory features. Through recursive feature elimination (RFE) and feature importance analysis using SHAP, 19 key features were identified. These features included MV duration, pressure support ventilation (PSV) level, spontaneous breathing trial (SBT) success time, heart rate, and other vital signs. After hyperparameter optimization using an automated toolkit, the model achieved a high predictive performance with an AUROC of 0.835 in internal validation and 0.803 in prospective validation, confirming its robustness and generalizability.

This study stands out for several reasons. First, it presented a large sample size: Unlike previous research limited to hundreds of patients, this study utilized a large-scale dataset of 16,189 patients, reducing the risk of overfitting and improving generalizability. The author also performed external validation, where the model was prospectively validated in a Cardiac Surgical ICU at Zhongshan Hospital, demonstrating its effectiveness in a real-world setting. Furthermore, this work employed feature importance analysis: By using SHAP values, the study identified the most significant predictors, such as MV duration and PSV level, offering interpretable and actionable insights for clinicians. Finally, the proposed approach presented high clinical usability with a web-based tool, allowing clinicians to input key variables and predict EF risk in real time, enhancing its applicability in clinical practice.

Despite its strengths, the study on EF prediction in [59] still face certain challenges. The study identified 19 key features using recursive feature elimination (RFE). However, other important variables, such as evaluations of diaphragm function, central venous to arterial PCO₂ difference, and cuff leak tests, were not included due to data unavailability in the MIMIC-IV database. These novel parameters, while clinically relevant, require further integration in future studies. Furthermore, scoring systems such as APACHE-II and SOFA were excluded to improve clinical usability. While this decision simplifies model implementation, these scores integrate a range of critical clinical parameters that may enhance

prediction accuracy.

Another recent study by Zhu et al. [60] was conducted to develop and validate machine learning models for predicting hospital mortality using features available within the first 24 hours of ICU admission. The study utilized the MIMIC-III database, a publicly available critical care database, and focused on combining widely used disease severity scores such as SAPS II, SOFA, and Oxford Acute Severity of Illness Score (OASIS) with other clinical and laboratory features to create more accurate predictive models.

The study also involved a large sample size with retrospective cohort of 25,659 adult ICU patients who underwent mechanical ventilation. A total of 66 predictors were extracted, including demographic information (age and gender), comorbidities (hypertension, diabetes, chronic heart failure, stroke, etc.), disease severity scores, vital signs (heart rate, blood pressure, temperature), and laboratory results (lactate, hemoglobin, glucose, creatinine, and white blood cell counts). Hospital mortality was defined as the primary outcome. To predict mortality, seven machine learning algorithms were implemented and compared: kNN, logistic regression, decision tree, random forest, bagging, neural network, and XGBoost. The dataset was randomly split into a training set (70%) and a testing set (30%), with model performance evaluated based on the area under the receiver operating characteristic curve (AUC) and calibration plots.

Similar to [59], feature importance was assessed using SHAP to provide insights into the most influential predictors. The results demonstrated that the XGBoost model achieved the best predictive performance among the seven models, with an AUC of 0.821 on the testing set. The random forest model followed closely with an AUC of 0.819. In contrast, the neural network underperformed compared to the other models, and its calibration curve showed suboptimal results. The SHAP analysis revealed that age, respiratory dysfunction, SAPS II score, maximum hemoglobin, and minimum lactate were the top five predictors of mortality. These findings underscore the importance of combining disease severity scores with additional clinical and laboratory data to improve mortality prediction for mechanically ventilated patients.

Despite its success, the study had several limitations. First, the models were developed using a single-center database (MIMIC-III), which may limit the generalizability of the findings to other ICUs or patient populations. External validation in multi-center datasets is necessary to confirm the robustness of the models. Second, the study only focused on hospital mortality as the outcome,

without considering other important outcomes such as 28-day mortality, ventilator-free days, or long-term survival. Additionally, variables with more than 30% missing values were excluded, and mean imputation was used for the remaining missing data, which may introduce bias or reduce the predictive power of the models. The inclusion of patients withdrawn from care could also affect the results, as this group may skew mortality predictions.

In summary, this study demonstrated the potential of machine learning models, particularly XGBoost, to predict mortality in mechanically ventilated ICU patients using early-stage clinical and laboratory features. By combining disease severity scores with additional predictors, the study improved upon traditional mortality prediction methods. However, the limitations, including the lack of external validation and exclusion of long-term outcomes, highlight the need for further studies to enhance the model's robustness and clinical applicability in broader settings.

a. Hospital Mortality in MV patients with HF Using ML Approaches

This section focuses on the key literature directly related to the field that this work aims to address. Recent work has focused on developing and validating machine learning models to predict hospital mortality in high-risk patients, particularly those with HF requiring mechanical ventilation [20], [61], [62]. In a recent study by Li et al. [17] specializing in CHF, a prediction model based on machine learning algorithms was created to assess hospital mortality in mechanically ventilated CHF patients. The model employed Least Absolute Shrinkage and Selection Operator (LASSO) regression for key feature selection and optimized model performance through hyperparameter tuning. Among the 11 ML algorithms tested, the CatBoost model exhibited the highest predictive accuracy, achieving an area under the receiver operating characteristic curve (AUC) of 0.833. After hyperparameter optimization, the model showed improved performance (AUC increased from 0.805 to 0.821) and was validated with an external dataset (AUC = 0.806). Prediction results were represented in a nomogram for clinical usability. This study demonstrated the effectiveness of CatBoost and key feature selection in improving prediction performance, though limitations included missing clinical variables (e.g., left ventricular ejection fraction) and a lack of differentiation between CHF subtypes (reduced or preserved ejection fraction). Additionally, the authors noted the need for prospective studies to confirm findings, as this study was retrospective, and they did not apply deep learning algorithms, which could further enhance predictive power.

In this section, we outline the scope of this work and review relevant background

and related studies. The next section, Methodology, will detail our approach to the problem and the techniques we use.

CHAPTER 3. METHODOLOGY

Building on the context established in Chapter 2, this section will present the methodology of this work, including a schematic overview and a detailed discussion of subsequent stages.

3.1 Overview

This section provides an overview of the main stages of our methodology, as illustrated in Figure 3.1. First, data was collected from version 3.0 of the MIMIC-IV database [63], focusing specifically on patients with CHF who underwent MV during the first 24 hours of their ICU admission. Subsequently, we conducted data preprocessing, beginning with data cleaning. In the cleaning step, we removed duplicate patient records and selected up to three records per patient. Additionally, rows and columns with missing values exceeding specific thresholds were excluded. Specifically, a row was removed if the number of missing values exceeded 30%, and a column was dismissed if more than 50% of its values were missing.

$$N_{missing_values}^{row} > 30\%; N_{missing_values}^{column} > 50\%$$

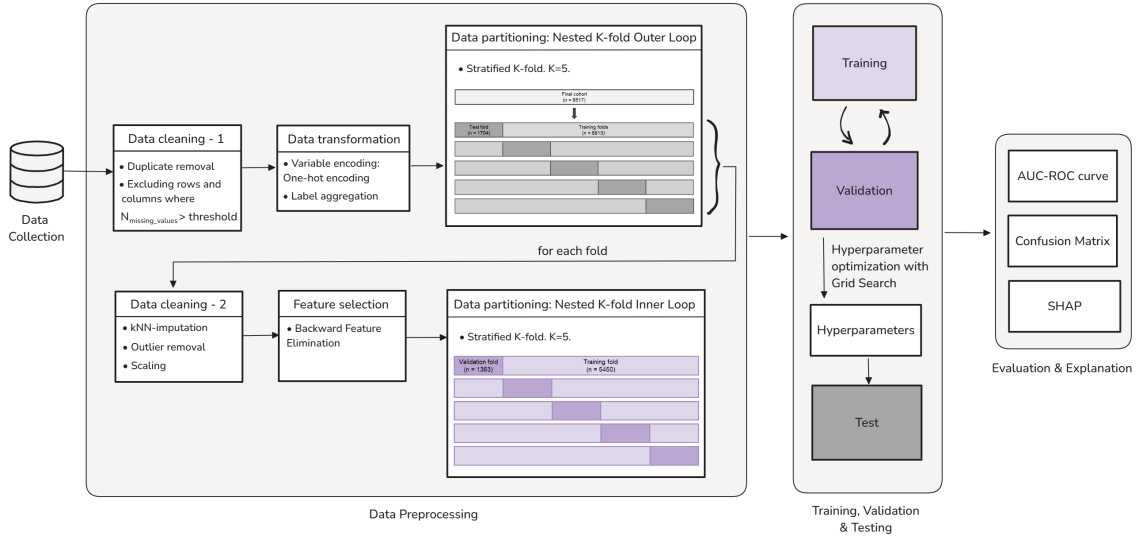


Figure 3.1: Schematic overview of the system in this work. Starting from collected data, preprocessing is then presented with data cleaning, data transformation, feature selection and a nested loop of data partitioning with stratified K-fold cross-validation [35] with $K = 5$. Next, we have training, validation and testing. Finally, evaluation and explanation with SHAP [43] are demonstrated.

After data cleaning, we performed data transformation using one-hot encoding for

the categorical variables. Additionally, we aggregated the values in the *race* column to reduce the number of race categories to six, ensuring consistency throughout the dataset. Next, we implemented data partitioning using a nested K-fold approach. In the outer loop, we applied stratified K-fold cross-validation [35] with $K = 5$, resulting in the separation of the test set from the rest of the dataset. For each fold, we conducted a second round of data cleaning, where missing data fields were imputed using the kNN algorithm. During this stage, we also removed outliers and scaled the dataset for normalization. The subsequent step involved feature selection, where backward feature elimination was employed. Following feature selection, the inner loop of data partitioning was conducted using the same stratified K-fold cross-validation with $K = 5$, generating separate training and validation folds.

The next phase involved training, validation, and testing. Training and validation were iterated across the folds to optimize hyperparameters using Grid Search [64]. Once the optimal hyperparameters were determined, the model was tested on the test set derived from the previous stage. After testing, the model was evaluated using the AUC-ROC curve and a confusion matrix to assess precision and recall. Finally, SHAP [43] was used to interpret the machine learning results as well as the relationship between key variables and patient outcome.

3.2 Data description

The MIMIC-IV [63] database is a publicly accessible, de-identified dataset containing electronic health record (EHR) data from the Beth Israel Deaconess Medical Center. This dataset is an essential tool for clinical research, supporting a wide range of studies, including those on predictive modeling, patient outcomes, and healthcare resource utilization.

This thesis utilizes the latest version, MIMIC-IV v3.0, which was released on July 19, 2024, and includes significant updates and expansions. This version incorporates patient stays recorded between 2020 and 2022, greatly increasing the dataset's scope and depth. A notable addition in v3.0 is the inclusion of out-of-hospital mortality data, allowing for the analysis of outcomes up to one year following a patient's discharge from the hospital or emergency department (ED). This enhancement is particularly valuable for research on long-term patient outcomes.

Key improvements in MIMIC-IV v3.0 compared to v2.2 are reflected in the increased data volume:

Patients: Increased from 299,712 to 364,627.

Admissions: Grew from 431,231 to 546,028.

ICU Stays: Expanded from 73,181 to 94,458.

These updates substantially enhance the dataset's representativeness and analytical power, enabling more detailed and large-scale investigations. The addition of recent data and mortality tracking supports research focusing on current clinical practices and patient care trends, ensuring the database remains highly relevant to modern healthcare challenges.

3.3 Data collection

MIMIC-IV v3.0 contains both ICD-9 and ICD-10 codes. To comprehensively capture data on CHF patients, I included all relevant ICD-9 and ICD-10 codes for CHF diagnoses. This approach ensures that patients diagnosed under either coding system are accounted for. Using these codes, 72 types of CHF were identified and extracted. The detailed summary of the ICD codes along with their corresponding titles is presented in Table 7.1

We started by selecting CHF patients who were diagnosed with CHF codes and required mechanical ventilation during their first ICU stay. To ensure data quality and relevance, we applied specific inclusion criteria to refine the dataset. These criteria included:

- Adults aged over 18 years
- CHF identified as the primary reason for ICU admission.
- Receipt of MV within the first 24 hours of ICU admission.

Relevant patient records were identified using ICD-9 and ICD-10 codes, which were then linked with demographic information, patient conditions, and ventilation-related data. After excluding incomplete and duplicate entries, we processed the dataset to arrive at a final cohort of 8,517 patient samples. The detailed process of patient selection is depicted in Fig 3.2

3.4 Feature selection and Data preprocessing

We initially selected 90 variables based on insights from literature review and expert recommendations. To refine this list, I excluded variables with more than 50% missing data, reducing the total number of variables to 70. Patients with more than 30% missing data were excluded from the analysis. It is important to note that all included patients had complete mortality information. This refinement reduced the number of eligible patients from 10,212 to 8,517.

The subject IDs and ICU stay IDs serve as the unique identifiers for patients and records of ICU admission, respectively. The selected features were categorized

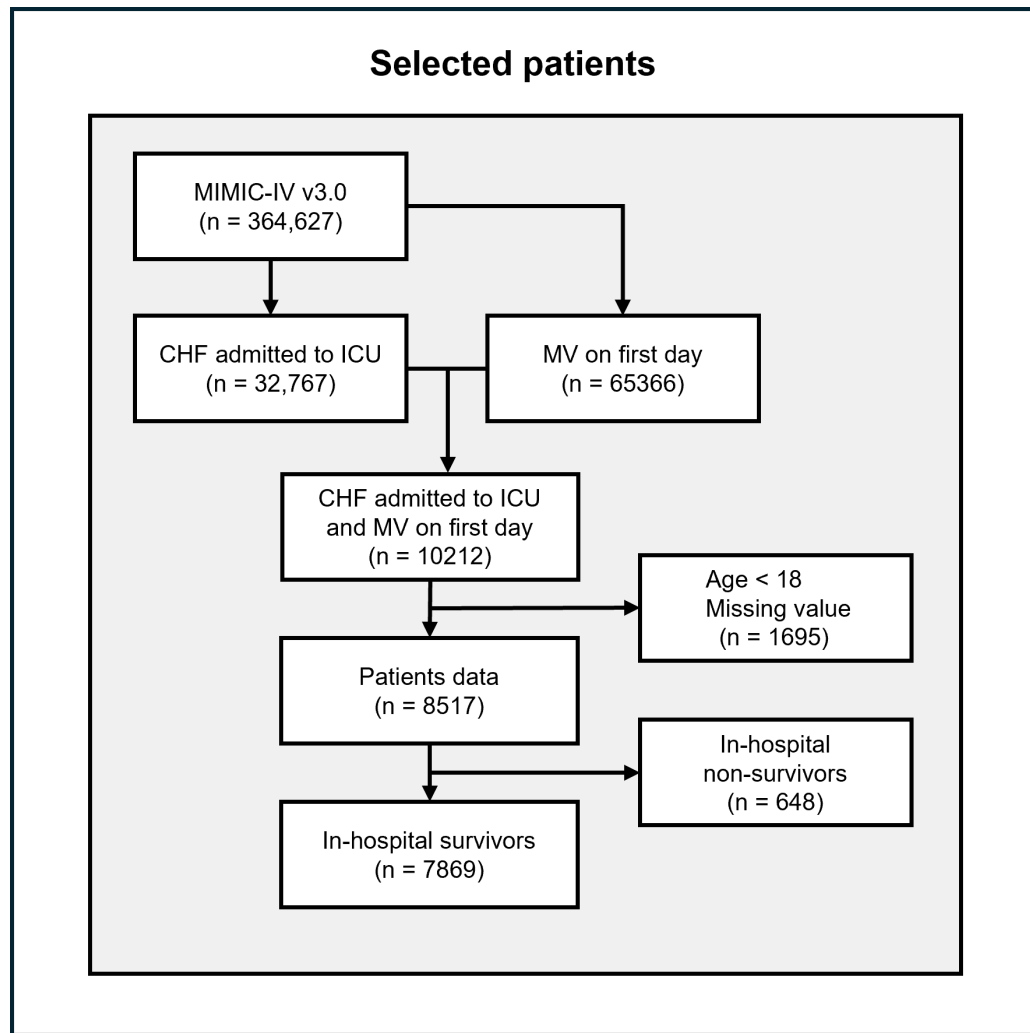


Figure 3.2: Flow diagram of the selection process of patients

into six groups: demographics, blood gas parameters, vital signs, laboratory results, comorbidities, and clinical scores. Variables removed due to excessive missing data are indicated in italic font. The variables included in each category are detailed in Table 3.1. The overall process for obtaining the final merged dataset is outlined in Fig 3.3

Category	Variables
Demographics	Age, Gender, Race, BMI (calculated from height and weight), GCS score, MV duration
Blood Gas	PaO ₂ , PaCO ₂ , SpO ₂ , PaO ₂ /FiO ₂ ratio, Total CO ₂ , Bicarbonate, Base excess, pH, Lactate, <i>Carboxyhemoglobin</i>
Vital Signs	Urine output, Heart rate, Respiratory rate, Temperature, <i>MAP(Mean Arterial Pressure, both systolic and diastolic)</i>

Category	Variables
Laboratory Results	NT-proBNP, CK-MB(Creatine kinase), Troponin T, Hematocrit, Hemoglobin, Platelet count, WBC, RBC, Anion gap, Creatinine, BUN, Glucose, Sodium, Calcium, Magnesium, Chloride, INR, PT, PTT, Basophils, Eosinophils, Lymphocytes, Monocytes, Neutrophils, ALT, ALP, AST, Total bilirubin, <i>Albumin, Bilirubin (direct and indirect), Globulin, Total Protein, GGT (Gamma-Glutamyl Transferase), LDH (Lactate Dehydrogenase), Bands, Metamyelocytes, Immature Granulocytes, NRBC (Nucleated Red Blood Cells), Atypical Lymphocytes, D-dimer, Fibrinogen, Thrombin, Amylase, CK-CPK</i>
Comorbidities	Sepsis, Hepatitis, Liver cirrhosis, Ventricular arrhythmia, Atrial fibrillation, Pneumonia, AMI (Acute Myocardial Infarction), OMI (Old Myocardial Infarction), Pleural effusion, Valvular disease, Hypertension, Anemia, CKD (Chronic Kidney Disease), Stroke, COPD (Chronic Obstructive Pulmonary Disease), Diabetes, AKI (Acute Kidney Injury), PE (Pulmonary Embolism), Angina pectoris
Clinical Scores	SOFA, SAPSII, LODS

Table 3.1: The variables included in the study, categorized into six groups: demographics, blood gas parameters, vital signs, laboratory results, comorbidities, and clinical scores. Each category highlights specific features relevant to assessing patient characteristics, physiological status, and disease severity during ICU admission.

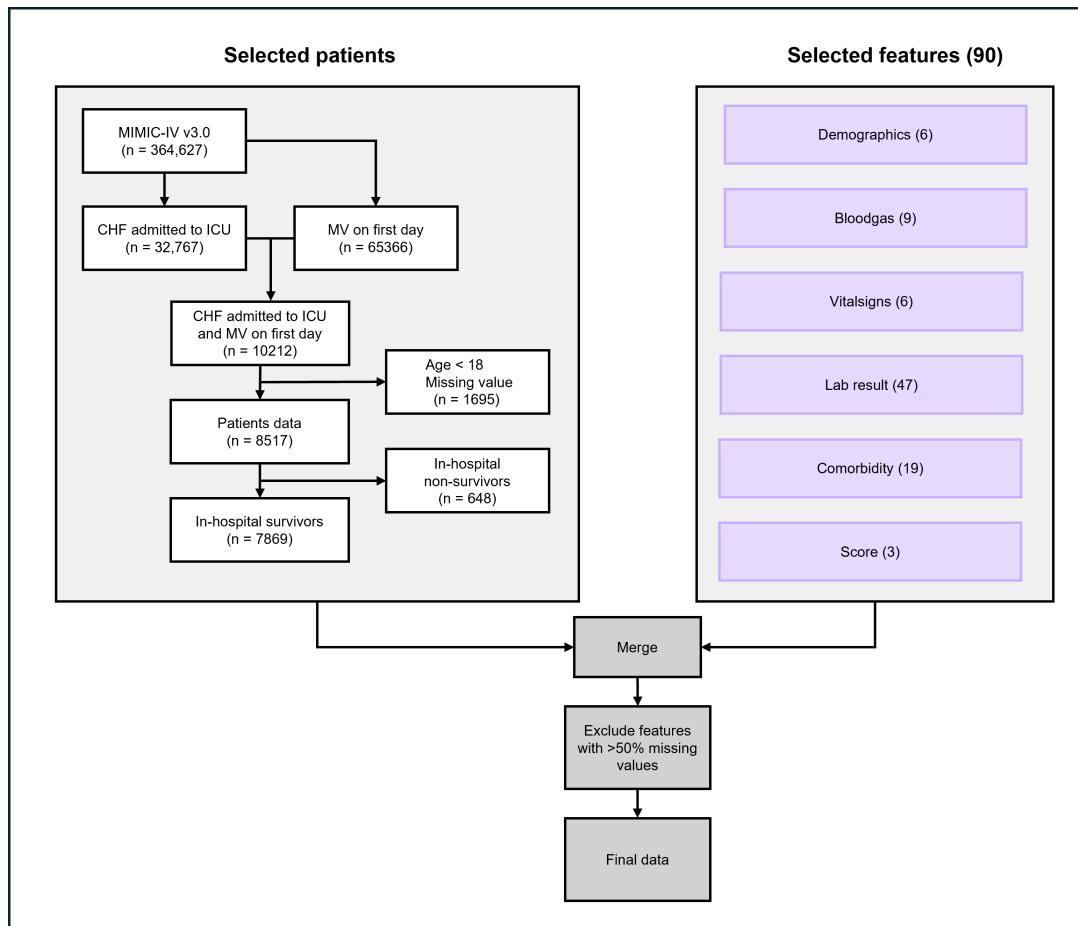


Figure 3.3: Flow diagram of data selection process

To prepare the data for analysis, categorical variables, such as gender and race, were one-hot encoded to create binary representations. For gender, the value 0 represented males, while 1 represented females. Patients were categorized into six racial groups: Asian, Black/African, White, Hispanic, American Indian, and Other. Each group was encoded as a binary variable, where a value of 1 indicated the patient belonged to the respective group, and 0 indicated otherwise. Similarly, comorbidities were encoded as binary variables, with 1 denoting the presence of the condition and 0 indicating its absence. Following this step, the dataset expanded to include a total of 75 features.

For patients with multiple ICU admissions, only data from the first admission was retained, ensuring consistency and focusing on the initial critical care intervention. Additionally, during the first ICU stay, variables with multiple recorded values were processed using three approaches: (1) capturing the first recorded value to reflect the initial condition upon admission, (2) retaining the last recorded value to represent the most recent status before discharge or death, and (3) calculating the mean of all recorded values to provide an overall average. These methods offered a comprehensive view of temporal data trends and enabled a nuanced understanding

of the progression of critical conditions and their impact on patient outcomes.

To ensure an unbiased evaluation of the model and optimize hyperparameters effectively, nested k-fold cross-validation was employed. A key preprocessing step involved KNN imputation to handle missing values. To prevent data leakage during preprocessing, a unique imputation object was created for each outer fold. This object was fitted only on the inner training set and then used to transform both the inner training and validation sets within the corresponding fold. This approach ensures that information from the outer test set does not influence the imputation process, thereby preserving the integrity of the evaluation and preventing overfitting. By isolating the test set from the training process, this method improves the robustness and generalizability of the model.

Following imputation, outliers were identified and removed to reduce noise and mitigate the impact of extreme values that could distort model performance. To standardize the data, features were scaled using the StandardScaler, centering the data around a zero mean and scaling it to unit variance, ensuring comparability across variables with different ranges and units.

For feature selection, Backward Feature Elimination, a Sequential Feature Selection technique, was employed to identify the most significant predictors. This process was further refined through an extensive review of existing literature and consultation with clinical experts. Ultimately, this rigorous approach resulted in the selection of 60 key features that were both statistically and clinically relevant, forming the foundation for model development and analysis.

Fig 3.4 demonstrates a horizontal bar chart showing the changes in number of features over the selection progress.

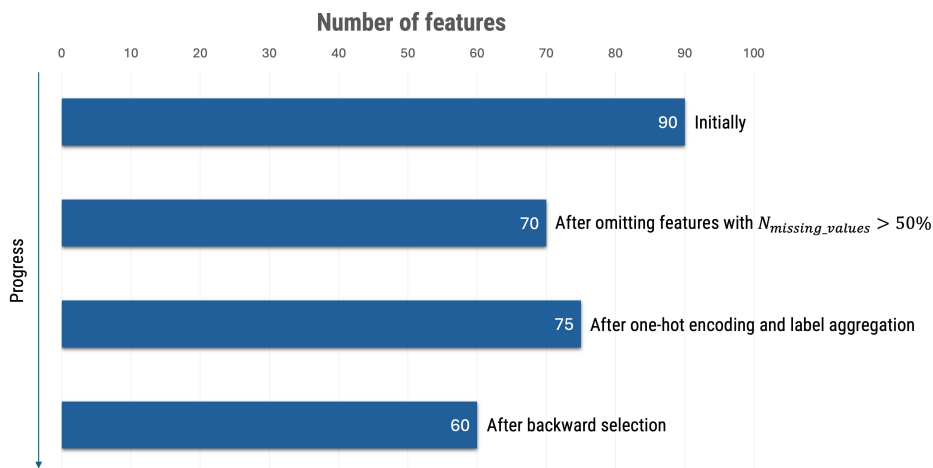


Figure 3.4: Demonstration of the changes in number of features over the selection progress.

3.5 Statistical analysis

To compare the differences between two groups of patients, survivors and non-survivors, a statistical analysis was performed for both continuous and categorical variables. Continuous variables were analyzed by calculating mean values and standard deviations (SD) for each group, and the differences were assessed using a two-sided t-test. The significance level was set at $P < 0.05$, where a p-value below this threshold indicates that the observed differences between the two groups are statistically significant and unlikely to be due to random chance.

For categorical variables, the frequency and percentage of occurrences in each group were calculated. A Chi-squared test was conducted to evaluate the association between categorical variables and the target outcome. Variables showing significant differences between the two groups ($P < 0.05$) suggest a stronger association with mortality.

This analysis highlights key variables that differ significantly between survivors and non-survivors, providing valuable insights into factors that may influence patient outcomes.

3.6 Model development and optimization

Various machine learning algorithms were developed and assessed, including Logistic Regression, CatBoost, and XGBoost. Additionally, advanced tabular deep learning models, such as TabNet, NODE, FT-Transformer, and GATE, were included to leverage their potential for capturing complex feature interactions and improving predictive accuracy. This comprehensive approach ensured that both traditional and state-of-the-art methods were thoroughly evaluated for their effectiveness in addressing the problem at hand.

For each model, nested k-fold cross-validation was implemented to ensure robust evaluation and effective hyperparameter optimization. The inner loop was used to fine-tune hyperparameters within a search space, enabling an initial assessment of each model's predictive performance under optimized conditions. After developing seven optimized models, their performance was tested on a separate test set to determine the algorithm that exhibited the best predictive capabilities for subsequent analyses.

The evaluation of these models focused primarily on the AUROC score, while AUROC 95% CI, accuracy and F1 scores were also computed to provide a more comprehensive performance assessment. AUROC was chosen as the primary metric due to its prevalent use in related research and its effectiveness in assessing model

performance [65].

After identifying the best model based on the AUROC results, a compact model was developed using an appropriate feature selection strategy. A grid search was then conducted on this compact model to identify the optimal configuration within a reduced hyperparameter search space. This focused approach leveraged the insights gained from the prior hyperparameter tuning process, allowing the search to concentrate on the most impactful parameters. The compact model, with the optimal combination of hyperparameters, was compared to the full model to demonstrate the effectiveness of the optimization process

3.7 Features importance

The impact of features on prediction scores was analyzed using SHAP values, a widely recognized method for evaluating feature importance. In the medical domain, SHAP values are particularly useful as they provide transparent and interpretable insights into model behavior, enabling healthcare professionals to better understand the reasoning behind predictive decisions [66], [67]. Positive SHAP values indicate that a feature contributes to an increased likelihood of the target outcome, while negative SHAP values reflect a reduction in this likelihood. The magnitude of SHAP values reflects the strength of a feature's influence on predictions, with larger absolute values indicating more significant contributions. By ranking features according to their impact, SHAP values pinpoint the most influential variables shaping the model's output, offering a comprehensive view of their role in predictions and the relationships between critical variables and patient outcomes [68].

3.8 Implementation

The experiments in this work were conducted with Intel-based CPU and an NVIDIA GeForce RTX 2080 Ti GPU. The implementation utilized Python as the primary programming language, with additional libraries including pandas for data manipulation, duckdb and sqlite3 for database management, pytorch_tabular for tabular deep learning model training and evaluation, sklearn for machine learning utilities and metrics, and matplotlib for visualization. This comprehensive combination of tools enabled efficient data processing, robust model training, and effective performance evaluation.

CHAPTER 4. RESULTS AND EVALUATION

In this section, we present the experimental results of the proposed methodology outlined in Chapter 3. We evaluate different approaches by comparing their accuracy and performance.

4.1 Baseline Characteristics

The final cohort consists of 8,517 unique stay IDs. In the initial step, nested k-fold cross-validation was applied to divide the dataset into 5 outer folds, ensuring robust evaluation and minimizing the risk of overfitting or underfitting. This process resulted in 6,813 records allocated to the outer training set and 1,704 records to the outer test set for each fold. Subsequently, for each outer training set, another k-fold cross-validation was performed, splitting it further into 5 inner training and validation sets. This inner division produced 5,450 records for the inner training set and 1,363 records for the validation set per fold. The detailed structure of this process is illustrated in Fig 4.1

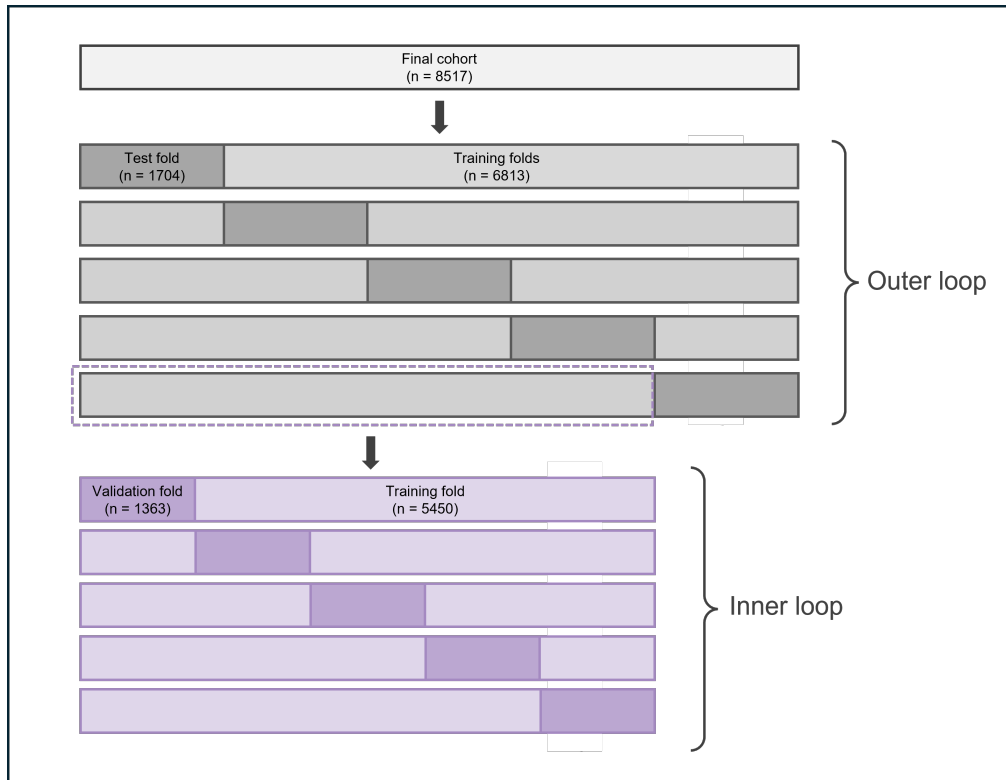


Figure 4.1: Process of splitting data using Nested k-Fold Cross-Validation

As shown in Table 4.1, the comparison of baseline characteristics highlights significant differences between in-hospital survivors and non-survivors in both the

training and test sets.

Variables	Training set			Test set		
	Survivors (<i>n</i> = 6295)	Non-survivors (<i>n</i> = 518)	<i>P</i> -value	Survivors (<i>n</i> = 1574)	Non-survivors (<i>n</i> = 130)	<i>P</i> -value
Demographics						
Female	2615 (41.54%)	243 (46.91%)	<0.05	647 (41.11%)	47 (36.15%)	0.312
Race:AMERICAN INDIAN	19 (0.30%)	2 (0.39%)	1.000	8 (0.51%)	0 (0.00%)	0.883
Race:ASIAN	181 (2.88%)	6 (1.16%)	<0.05	37 (2.35%)	3 (2.31%)	1.000
Race:BLACK/AFRICAN	841 (13.36%)	20 (3.86%)	<0.05	188 (11.94%)	1 (0.77%)	<0.05
Race:HISPANIC	226 (3.59%)	9 (1.74%)	<0.05	62 (3.94%)	2 (1.54%)	0.253
Race:WHITE	4078 (64.78%)	262 (50.58%)	<0.05	1024 (65.06%)	65 (50.00%)	<0.05
Race:OTHER	950 (15.09%)	219 (42.28%)	<0.05	255 (16.20%)	59 (45.38%)	<0.05
Age	70.31 ± 13.30	75.48 ± 12.60	<0.05	70.62 ± 12.74	73.85 ± 13.40	<0.05
Bmi	31.46 ± 79.57	28.50 ± 7.53	0.475	29.76 ± 8.47	30.19 ± 11.18	0.654
MV Duration	52.57 ± 86.57	90.63 ± 111.62	<0.05	55.67 ± 97.60	78.88 ± 93.94	<0.05
Gcs score	13.22 ± 3.37	9.83 ± 3.87	<0.05	13.31 ± 3.32	9.92 ± 3.77	<0.05
Blood gas						
PaO ₂	224.44 ± 139.13	170.53 ± 118.12	<0.05	228.13 ± 138.32	155.50 ± 115.36	<0.05
PaCO ₂	45.83 ± 15.85	45.34 ± 16.76	0.523	45.92 ± 16.87	43.86 ± 14.97	0.199
SpO ₂	97.51 ± 4.56	95.91 ± 7.77	<0.05	97.76 ± 4.40	96.07 ± 6.28	<0.05
PaO ₂ /FiO ₂ ratio	241.12 ± 165.66	211.95 ± 143.32	<0.05	244.57 ± 173.06	209.60 ± 153.58	<0.05
Total CO ₂	26.77 ± 7.08	24.45 ± 7.71	<0.05	26.59 ± 6.42	23.52 ± 6.51	<0.05
Bicarbonate	24.52 ± 5.85	22.50 ± 6.41	<0.05	24.41 ± 5.67	21.64 ± 6.00	<0.05
Base excess	-0.23 ± 6.28	-3.14 ± 7.66	<0.05	-0.32 ± 5.75	-3.88 ± 6.39	<0.05
PH	7.35 ± 0.11	7.31 ± 0.14	<0.05	7.35 ± 0.11	7.30 ± 0.13	<0.05
Laboratory results						
NT-proBNP	9713.19 ± 12076.33	13096.14 ± 15980.54	<0.05	10544.19 ± 12352.45	11894.54 ± 11521.37	0.603
CK-MB	15.83 ± 44.96	32.74 ± 86.60	<0.05	16.97 ± 50.12	39.26 ± 91.79	<0.05
Troponin T	0.53 ± 1.69	0.88 ± 2.57	<0.05	0.59 ± 2.42	1.38 ± 3.08	<0.05
Hematocrit	32.90 ± 7.19	34.58 ± 7.35	<0.05	32.94 ± 6.97	33.20 ± 7.34	0.682
Hemoglobin	10.56 ± 2.40	11.09 ± 2.40	<0.05	10.60 ± 2.34	10.60 ± 2.44	0.990
Platelet	223.78 ± 115.40	209.72 ± 109.74	<0.05	221.05 ± 114.62	202.48 ± 112.36	0.075
WBC	12.01 ± 7.54	14.79 ± 10.71	<0.05	11.93 ± 9.76	14.10 ± 5.58	<0.05
RBC	3.59 ± 0.83	3.71 ± 0.83	<0.05	3.61 ± 0.81	3.59 ± 0.81	0.857
Anion gap	15.33 ± 4.87	17.76 ± 5.48	<0.05	15.39 ± 4.90	18.12 ± 6.17	<0.05
Creatinine	1.82 ± 1.77	1.90 ± 1.54	0.299	1.86 ± 1.79	1.91 ± 1.42	0.734
BUN	34.33 ± 25.13	40.44 ± 29.24	<0.05	35.38 ± 26.63	42.44 ± 33.05	<0.05
Lactate	2.20 ± 2.18	3.54 ± 3.13	<0.05	2.16 ± 2.07	3.66 ± 3.33	<0.05
Glucose	153.50 ± 85.01	168.59 ± 89.72	<0.05	157.46 ± 103.19	174.48 ± 97.44	0.070
Sodium	138.35 ± 5.06	138.40 ± 6.16	0.842	138.26 ± 5.41	139.42 ± 6.31	<0.05
Potassium	4.45 ± 0.87	4.48 ± 0.93	0.527	4.48 ± 0.90	4.38 ± 0.88	0.222
Calcium	8.57 ± 0.88	8.34 ± 1.02	<0.05	8.56 ± 0.93	8.33 ± 1.01	<0.05
Magnesium	2.12 ± 0.57	2.11 ± 0.47	0.729	2.12 ± 0.46	2.15 ± 0.40	0.521
Chloride	101.73 ± 6.91	101.49 ± 7.22	0.433	101.75 ± 7.21	102.59 ± 6.98	0.197
INR	1.63 ± 1.15	1.86 ± 1.56	<0.05	1.60 ± 1.26	1.96 ± 1.15	<0.05
PT	17.68 ± 11.68	20.01 ± 15.87	<0.05	17.36 ± 12.02	21.07 ± 11.99	<0.05
PTT	40.81 ± 24.55	46.73 ± 31.92	<0.05	40.27 ± 24.34	46.95 ± 32.56	<0.05
Basophils	0.04 ± 0.07	0.03 ± 0.06	0.130	0.04 ± 0.04	0.04 ± 0.04	0.734
Eosinophils	0.13 ± 0.23	0.07 ± 0.12	<0.05	0.13 ± 0.27	0.07 ± 0.16	<0.05

Variables	Training set			Test set		
	Survivors (<i>n</i> = 6295)	Non-survivors (<i>n</i> = 518)	<i>P</i> -value	Survivors (<i>n</i> = 1574)	Non-survivors (<i>n</i> = 130)	<i>P</i> -value
Lymphocytes	1.65 ± 4.14	1.89 ± 6.92	0.294	1.67 ± 2.89	1.09 ± 0.64	<0.05
Monocytes	0.67 ± 0.57	0.86 ± 2.14	<0.05	0.68 ± 0.81	0.80 ± 0.57	0.134
Neutrophils	10.86 ± 6.58	12.60 ± 7.87	<0.05	10.80 ± 8.45	12.44 ± 5.44	<0.05
ALT	116.15 ± 465.50	224.96 ± 715.88	<0.05	79.36 ± 212.65	256.14 ± 813.33	<0.05
ALP	120.28 ± 123.64	119.13 ± 101.48	0.858	115.49 ± 92.11	125.59 ± 123.85	0.313
AST	186.21 ± 755.49	346.31 ± 1006.82	<0.05	126.75 ± 364.79	598.57 ± 2721.50	<0.05
Total bilirubin	1.12 ± 1.94	1.82 ± 4.37	<0.05	1.01 ± 1.61	2.40 ± 4.68	<0.05
Scores						
SOFA	2.91 ± 2.64	3.26 ± 3.01	<0.05	2.88 ± 2.62	3.62 ± 3.19	<0.05
SAPS-II	44.66 ± 14.06	55.58 ± 15.81	<0.05	44.44 ± 13.44	55.51 ± 16.86	<0.05
LODS	6.49 ± 3.06	8.58 ± 3.33	<0.05	6.48 ± 2.96	8.74 ± 3.48	<0.05
Vital signs						
Urine Output	1679.04 ± 1274.01	1175.95 ± 1292.40	<0.05	1676.93 ± 1196.88	1048.22 ± 1055.28	<0.05
Heart rate	87.70 ± 19.74	91.37 ± 21.12	<0.05	87.05 ± 19.33	94.60 ± 23.86	<0.05
Respiratory rate	19.46 ± 6.39	21.27 ± 6.80	<0.05	19.25 ± 6.14	22.17 ± 6.17	<0.05
Temperature	36.59 ± 0.93	36.50 ± 1.21	0.057	36.56 ± 0.94	36.48 ± 1.16	0.400
Comorbidity						
Sepsis	4784 (76.00%)	392 (75.68%)	0.912	1178 (74.84%)	102 (78.46%)	0.417
Hepatitis	315 (5.00%)	34 (6.56%)	0.149	78 (4.96%)	12 (9.23%)	0.059
Liver cirrhosis	296 (4.70%)	37 (7.14%)	<0.05	72 (4.57%)	14 (10.77%)	<0.05
Ventricular arrhythmia	704 (11.18%)	76 (14.67%)	<0.05	185 (11.75%)	29 (22.31%)	<0.05
Atrial fibrillation	3205 (50.91%)	295 (56.95%)	<0.05	813 (51.65%)	67 (51.54%)	1.000
Pneumonia	1841 (29.25%)	197 (38.03%)	<0.05	438 (27.83%)	49 (37.69%)	<0.05
AMI	127 (2.02%)	22 (4.25%)	<0.05	28 (1.78%)	6 (4.62%)	0.058
OMI	1003 (15.93%)	72 (13.90%)	0.247	241 (15.31%)	14 (10.77%)	0.205
Pleural effusion	557 (8.85%)	65 (12.55%)	<0.05	140 (8.89%)	14 (10.77%)	0.577
Valvular disease	2409 (38.27%)	156 (30.12%)	<0.05	614 (39.01%)	39 (30.00%)	0.053
Hypertension	3663 (58.19%)	278 (53.67%)	0.050	932 (59.21%)	69 (53.08%)	0.203
Anemia	3630 (57.66%)	243 (46.91%)	<0.05	917 (58.26%)	54 (41.54%)	<0.05
CKD	2483 (39.44%)	180 (34.75%)	<0.05	648 (41.17%)	45 (34.62%)	0.171
Stroke	356 (5.66%)	99 (19.11%)	<0.05	87 (5.53%)	23 (17.69%)	<0.05
COPD	1015 (16.12%)	82 (15.83%)	0.910	239 (15.18%)	21 (16.15%)	0.866
Diabetes	2810 (44.64%)	182 (35.14%)	<0.05	731 (46.44%)	47 (36.15%)	<0.05
AKI	2934 (46.61%)	343 (66.22%)	<0.05	739 (46.95%)	86 (66.15%)	<0.05
PE	357 (5.67%)	26 (5.02%)	0.603	102 (6.48%)	12 (9.23%)	0.306
Angina pectoris	374 (5.94%)	15 (2.90%)	<0.05	89 (5.65%)	5 (3.85%)	0.504

Table 4.1: Baseline characteristics of the training set and test set

The statistical analysis identified several significant differences between survivors and non-survivors. Non-survivors had notably longer durations on MV, with durations varying widely, ranging from over 200 hours to only a few hours before death. They also exhibited higher heart and respiratory rates, indicating greater physiological stress. Elevated levels of CK-MB were observed in non-survivors,

aligning with findings from other studies [69], [70], and higher blood urea nitrogen (BUN) levels, which are strongly associated with poor prognosis in heart failure patients, as demonstrated in the study by Siyu Duan et al [71]. Non-survivors also experienced a higher rate of stroke and had elevated white blood cell (WBC) counts, reflecting systemic inflammation or infection, a pattern corroborated by findings in other research [72]–[74]. These results underscore critical physiological and clinical factors linked to poorer outcomes in heart failure patients.

4.2 Models Development and Evaluation

To provide a thorough comparison, seven models were included, consisting of three widely used machine learning algorithms—Logistic Regression, XGBoost, and CatBoost—selected for their frequent use in the literature, alongside four advanced tabular deep learning models: TabNet, NODE, FT-Transformer, and GATE. This comprehensive selection allowed me to rigorously evaluate the effectiveness of data preprocessing, imputation, and feature selection methods. Logistic Regression, a fundamental algorithm, calculates binary outcome probabilities using a logistic function, making it a common baseline in predictive modeling. XGBoost, an efficient implementation of gradient boosting, iteratively improves predictive accuracy by optimizing decision trees. CatBoost, designed to handle categorical data efficiently, has been demonstrated in prior studies to outperform many models in hospital mortality prediction tasks, making it a strong contender in this comparison.

The deep learning models bring additional flexibility and capability to handle complex tabular data. TabNet, which leverages attention mechanisms, dynamically selects which features to attend to during each step of the learning process, making it both interpretable and highly performant. NODE (Neural Oblivious Decision Ensembles) represents another innovative approach to tabular data, blending decision tree concepts with neural networks to achieve superior accuracy while maintaining efficiency. FT-Transformer adapts the transformer architecture—originally designed for sequential data—to tabular datasets, utilizing multi-head attention to capture feature interactions effectively. Finally, GATE combines the strengths of decision trees and neural networks with its unique gated learning mechanism, providing robust performance and interpretability.

For each model, nested k-fold cross-validation was employed, allowing hyperparameters to be optimized within the inner loop using a simplified search space. This approach ensured an initial assessment of predictive performance under optimized conditions. The comparison of these seven models is summarized in Fig. 4.2, with detailed results presented in Table 4.2. The GATE model exhibited

the highest predictive performance among all models, achieving an impressive AUC of 0.878. CatBoost also demonstrated strong predictive capabilities, achieving an AUC of 0.867, which aligns with its widespread application in various studies [17], [20], [59], [62]. Both the FT-Transformer and Logistic Regression models also delivered commendable results, highlighting their reliability in predictive tasks. Based on these results, GATE was selected as the primary algorithm. This choice highlights GATE’s effectiveness in capturing intricate relationships within the data and underscores the potential of deep learning approaches in advancing mortality prediction tasks.

Models	AUROC	Accuracy	AUROC 95% CI
GATE	0.878	0.873	[0.8445 – 0.9068]
FT-Transformer	0.855	0.896	[0.8198 – 0.8862]
NODE	0.831	0.862	[0.7959 – 0.8617]
TabNet	0.800	0.873	[0.7583 – 0.8385]
CatBoost	0.867	0.897	[0.8327 – 0.8978]
XGBoost	0.854	0.895	[0.8174 – 0.8876]
Logistic Regression	0.858	0.897	[0.8229 – 0.8913]

Table 4.2: Evaluated seven different models: Logistic Regression, XGBoost, CatBoost, TabNet, NODE, FT-Transformer and GATE. The evaluation metrics included AUROC, AUROC 95% CI, and the accuracy score.

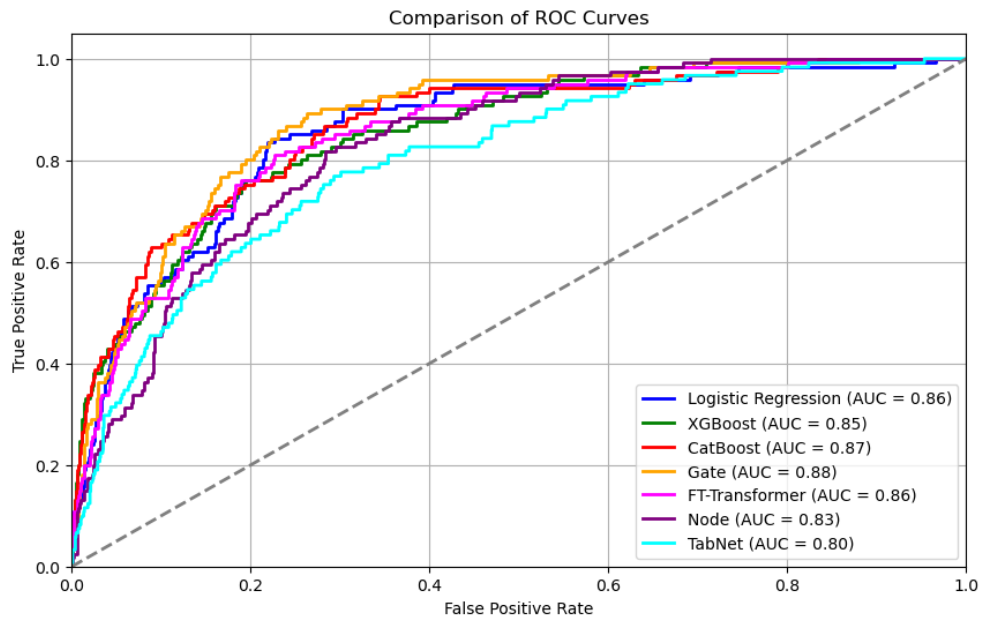


Figure 4.2: The AUROC curves of the seven models for the test set: Logistic Regression, XGBoost, CatBoost, TabNet, NODE, FT-Transformer and GATE

Next, we compared the performance of the GATE model with other state-of-the-art studies that have similarly focused on predicting mortality in CHF patients. This

evaluation underscores the GATE model's ability to capture critical features and deliver accurate predictions, demonstrating its potential as an advanced and reliable tool for assessing mortality risk in CHF patients. The results of this comparison are presented in Table 4.3.

Table 4.3: Comparison of AUROC Performance with State-of-the-Art Models

Study/Model	Model Used	AUROC
This Study	GATE	0.878
[20]	CatBoost	0.863
[62]	CatBoost	0.844
[75]	FT-Transformer	0.743

The chosen GATE model was trained using three distinct datasets, each reflecting a different aggregation approach: the first recorded value, the last recorded value, and the mean of all recorded values. This process resulted in three separate models. Additionally, an ensemble model was developed by combining the predictions of these three individual models, creating a fourth, comprehensive resemble model.

To evaluate the performance of these models, three corresponding test datasets were utilized, each aligned with the respective aggregation methods (first value, last value, and mean). This evaluation was conducted to determine which model offers the best suitability for specific clinical objectives, such as early prediction or risk assessment, both of which are critical for real-time decision-making in medical settings. Furthermore, the analysis investigated the robustness of each model in handling variations across datasets, providing key insights into their reliability and applicability in diverse real-world scenarios.

The results of the four models when tested with the three corresponding datasets are presented in Fig 4.3, Fig 4.4 and Fig 4.5 respectively.

Overall, the ensemble model consistently demonstrated strong performance across all three test cases, making it the preferred choice for further analysis and application.

4.3 Features Importance

The SHAP analysis conducted on the test set is visualized in Figure 4.6. Feature values are depicted using a gradient scale, with blue representing the lowest values. Positive SHAP values indicate an elevated likelihood of in-hospital mortality, while negative values suggest a reduced risk. Features are ranked by the magnitude of their SHAP values across all samples. The findings reveal that patient age and MV duration are the most influential factors in predicting in-hospital mortality within the

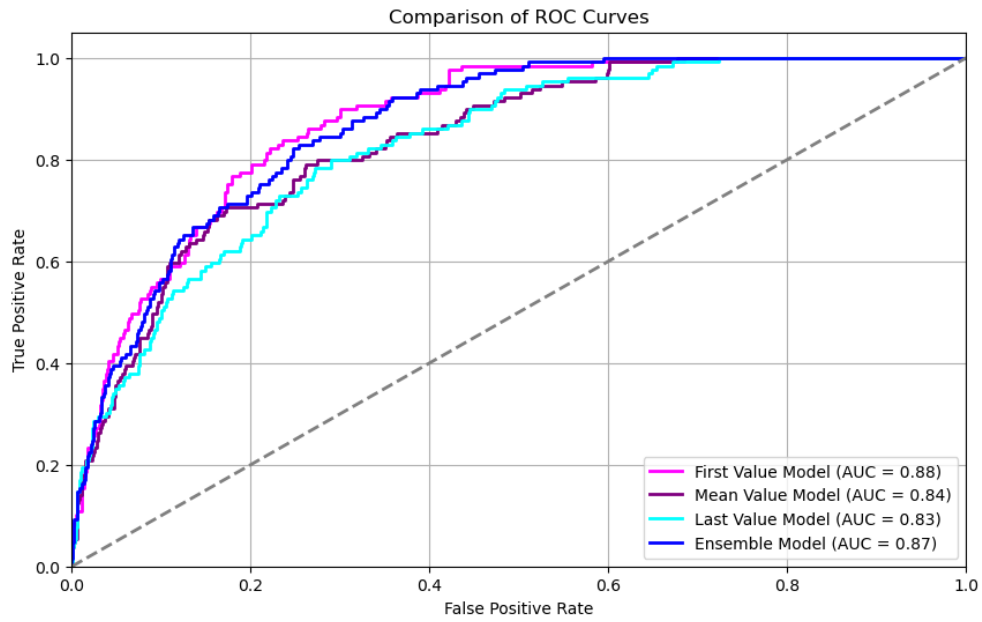


Figure 4.3: The AUROC curves of four models when testing with first value dataset

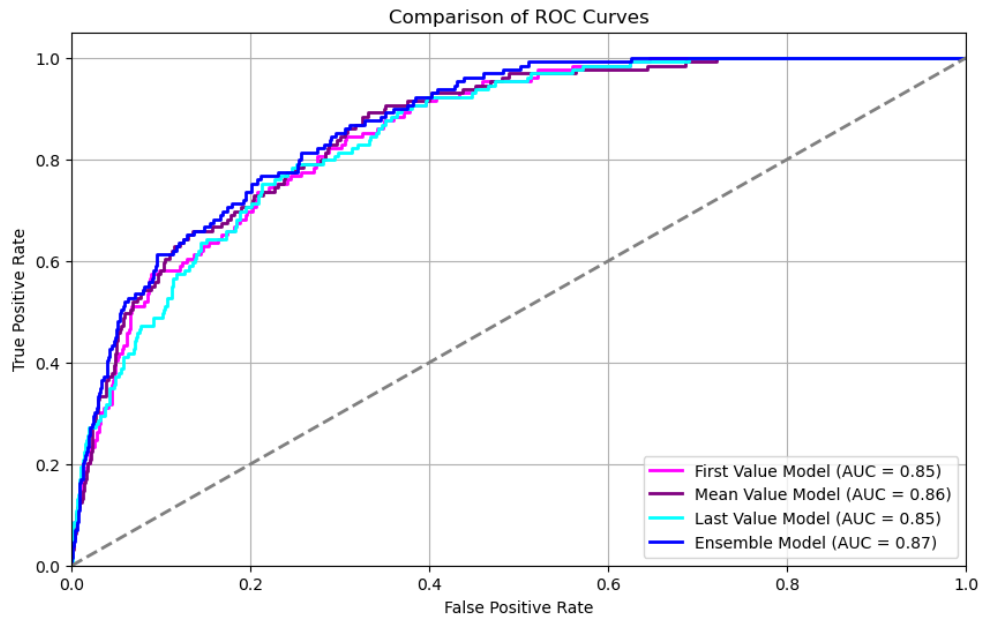


Figure 4.4: The AUROC curves of four models when testing with mean value dataset

final model. Specifically, prolonged MV durations and older are strongly correlated with a greater risk of mortality.

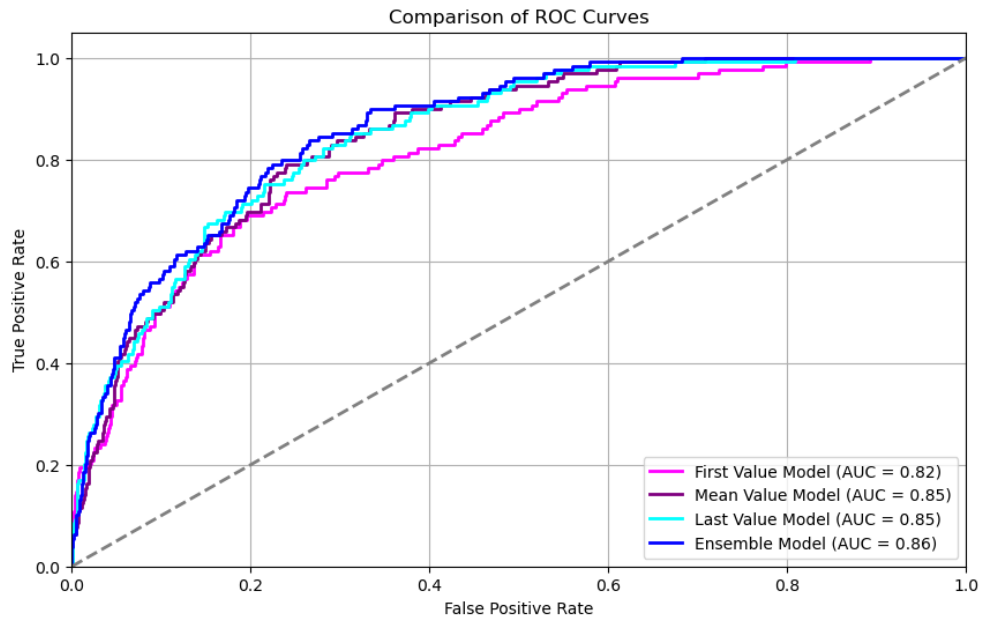


Figure 4.5: The AUROC curves of four models when testing with last value dataset

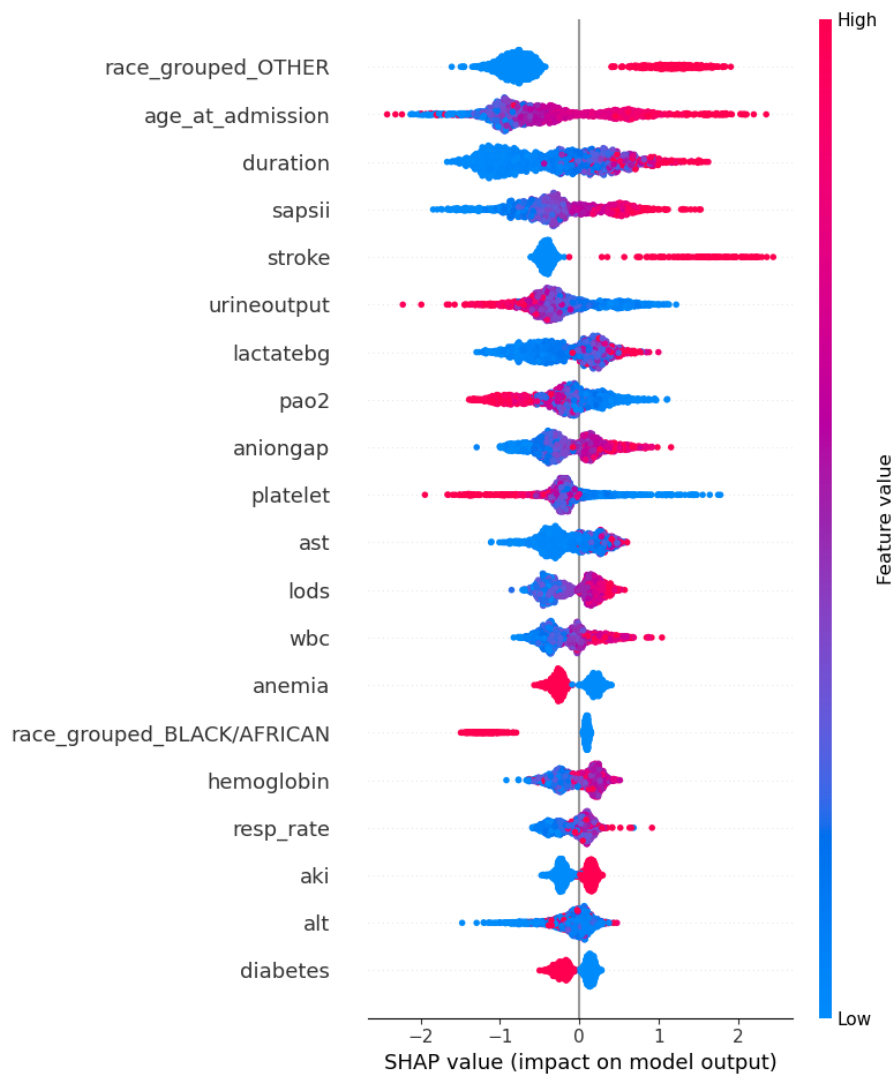


Figure 4.6: Distribution of the impacts important features had on the output of the model estimated using the SHAP values

Using SHAP analysis, key demographic and clinical features was identified, including variables such as race, MV duration, age, stroke, anemia, acute kidney injury (AKI), diabetes history, SAPSII and LODS scores, urine output, lactate levels, partial pressure of oxygen in arterial blood (PaO₂), anion gap, platelet count, AST and ALT levels, white blood cell count (WBC), hemoglobin levels and respiratory rate. These features were utilized to construct a compact model tailored to the most predictive and clinically relevant variables.

This compact model was then subjected to a thorough hyperparameter tuning process to optimize its performance. By focusing on this refined feature set, the compact model achieved a significant reduction in computational complexity while enhancing its interpretability.

Comparisons between the full model and the compact model, following hyperparameter optimization (HPO), were performed to evaluate the impact of the optimization process. The predictive performance of the compact model, optimized with 20 key features, showed a slight decrease compared to the full model, with AUCs of 0.86 and 0.88, respectively. This result is shown in Fig 4.7.

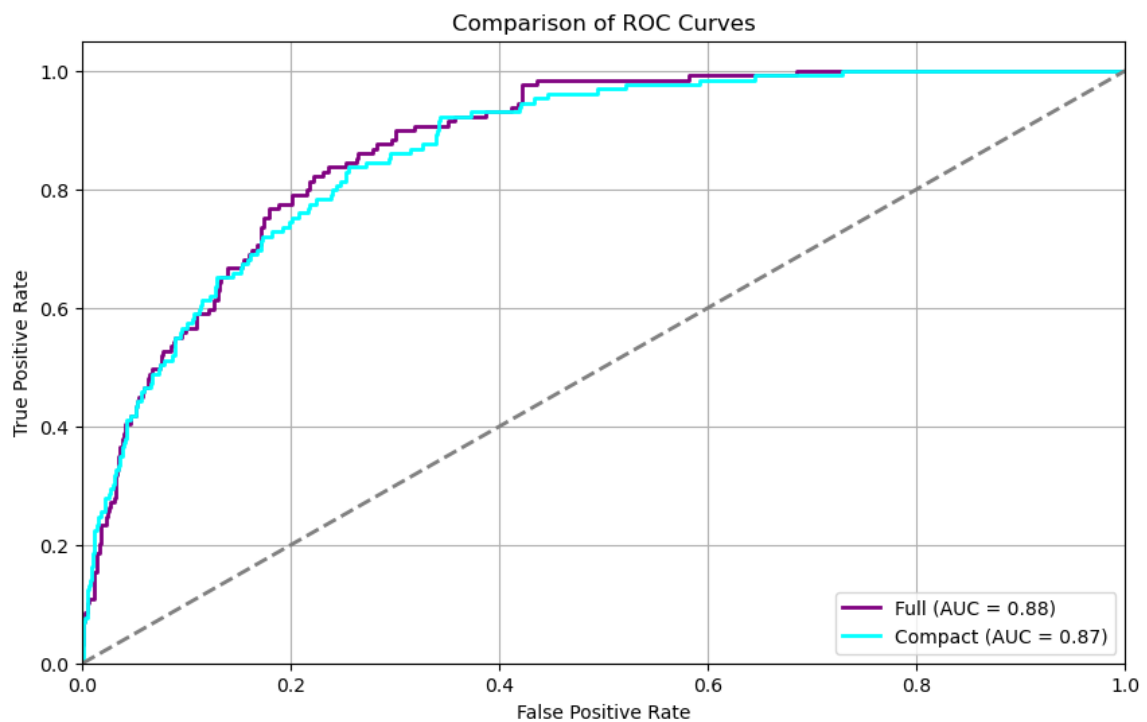


Figure 4.7: Comparison of the full and compact GATE models: The full model was built utilizing all available features, whereas the compact model was constructed using 20 key features identified through the SHAP algorithm. Both models underwent hyperparameter optimization to ensure their best possible performance.

As expected, the compact model exhibited a slight reduction in AUROC

compared to the full model. This result highlights that other features could be excluded without significantly compromising the model's predictive accuracy.

CHAPTER 5. DISCUSSION

In this study, we developed and validated a predictive model to estimate hospital mortality among mechanically ventilated patients with CHF. Data were extracted from the MIMIC database, initially filtering 90 variables, which were subsequently reduced to 60 through backward feature selection. Seven predictive models were evaluated, with the GATE deep learning model demonstrating superior performance. To further assess data aggregation methods, three GATE models were trained using different strategies: one with the first recorded values, another with the last recorded values, and a third with the mean of all values. The results indicated that an ensemble of these three models outperformed individual models, suggesting its robustness and generalizability for prediction tasks.

Using SHAP analysis, 20 critical features were identified, enhancing the model's usability compared to prior studies. As shown in Fig 4.6, these features include race, MV duration, age, history of stroke, anemia, acute kidney injury (AKI) and diabetes, SAPSII score, LODS score, urine output, lactate levels, partial pressure of oxygen in arterial blood (PaO₂), anion gap, platelet count, AST levels, ALT levels, white blood cell count (WBC), hemoglobin levels, and respiratory rate. Although the compact model constructed with these 20 key features showed a slight reduction in predictive performance compared to the full model, it maintained strong accuracy and demonstrated its practicality without significantly compromising the model's effectiveness.

This SHAP summary plot provides insights into the importance and impact of individual features on the predictions made by a machine learning model. Previous studies have consistently demonstrated that these features are significantly associated with an elevated risk of mortality in patients CHF. MV duration is a critical indicator of adverse outcomes, as extended use often reflects severe respiratory compromise and critical illness [17], [52], [61]. Longer MV duration is strongly associated with higher in-hospital mortality. Age is another well-established predictor, with older patients experiencing higher mortality rates due to age-related declines in physiological reserve. Elderly patients are more likely to have multiple comorbidities, as well as diminished cardiac and pulmonary functions, leading to a poorer prognosis compared to younger populations.[17], [20], [60]

Clinical comorbidities, including stroke, anemia, acute kidney injury (AKI), and diabetes [19], [76]–[79], are well-documented as factors that exacerbate complications in CHF and significantly increase the risk of mortality. Indicators of

perfusion and oxygenation, such as urine output, lactate levels, and arterial oxygen partial pressure (PaO₂), are vital for predicting patient outcomes. Higher urine output is associated with a decreased likelihood of in-hospital mortality, reflecting better renal function and overall organ health [80]. Conversely, elevated lactate levels indicate worse outcomes, highlighting metabolic stress and poor perfusion [60], [61]. Higher PaO₂ values are linked to reduced mortality risk, emphasizing the importance of effective oxygen delivery [60].

Metabolic and hematological markers also play crucial roles in prognosis. An elevated anion gap is associated with increased mortality risk [81], while lower platelet counts are linked to poor outcomes [82]. Low hemoglobin levels are another significant indicator, often correlating with worse survival [18], [19], [60], [61]. Higher white blood cell (WBC) counts, a marker of systemic inflammation, are positively correlated with increased mortality risk [19], [72]–[74]. Liver function markers, such as elevated aspartate aminotransferase (AST) and alanine aminotransferase (ALT) levels, reflect hepatic dysfunction and are linked to poorer outcomes [83], [84]. Additionally, respiratory rate serves as a key marker of systemic stress and organ compromise, further underscoring its clinical relevance.

Prognostic scoring systems such as the SAPSII and LODS [60], [61] are widely accepted tools for assessing disease severity and predicting patient outcomes. These systems integrate multiple variables to provide a comprehensive evaluation of organ dysfunction and mortality risk. As expected, higher scores in these systems are strongly associated with increased mortality.

Collectively, these features form a biologically and clinically robust framework for mortality prediction in CHF patients. Their inclusion in predictive models ensures clinical relevance and reflects the intricate interplay of physiological, metabolic, and inflammatory processes influencing patient outcomes. By leveraging this evidence base, predictive models can achieve greater accuracy and provide actionable insights for improving patient care.

CHAPTER 6. CONCLUSIONS

6.1 Summary

This thesis aims to develop a robust and reliable predictive model for in-hospital mortality among mechanically ventilated patients with congestive heart failure, or CHF. The primary goal was to create an accurate, interpretable, and efficient tool to assist clinicians in making informed decisions and improving patient outcomes. By leveraging advanced machine learning techniques and comprehensive feature analysis, this study bridges the gap between data-driven insights and practical clinical application.

Using data from the MIMIC-IV database, the research process began with filtering and statistical analysis to identify features relevant to patient status. After extensive preprocessing, a GATE model was trained to predict in-hospital mortality with high accuracy. Through SHAP analysis, the study identified 20 critical clinical and demographic variables that strongly influence survival outcomes. These variables included MV duration, age, SAPSII and LODS scores, markers of organ dysfunction, and other key indicators.

The identified features were then used to develop and optimize a compact predictive model. This model retained robust predictive power (AUROC of 0.87) while significantly reducing computational complexity and improving interpretability compared to the full model (AUROC of 0.88). The compact model's efficiency and focus on essential features make it a practical tool for real-time decision-making in clinical settings.

The findings of this thesis emphasize the importance of key clinical and demographic factors in determining survival outcomes for mechanically ventilated CHF patients. By highlighting these critical features, the study provides actionable insights to help healthcare professionals prioritize patient care and optimize treatment strategies. This research not only contributes to improving predictive modeling in critical care but also serves as a foundation for more targeted and effective patient management approaches.

While this thesis presents a robust predictive model for in-hospital mortality among mechanically ventilated CHF patients, several limitations should be acknowledged. Firstly, the training, validation, and test datasets used in this study were all sourced from the MIMIC-IV database. Although this database is comprehensive and widely used for critical care research, the lack of independent

datasets from different healthcare systems limits the generalizability of the model. Testing the model on external datasets would provide a more reliable assessment of its performance across diverse patient populations and clinical settings. Secondly, the study relied on static feature representation, such as first, last, and mean recorded values, to train the predictive models. However, critical care data often have a dynamic and temporal nature. Incorporating time-series analysis could capture trends, fluctuations, and other temporal patterns in the data, potentially improving the accuracy and clinical relevance of the model. This limitation suggests that future work should explore the integration of time-series methodologies to enhance predictive capabilities.

By addressing these limitations in future research, the model could achieve broader applicability and more nuanced predictions, further advancing its utility in clinical practice.

6.2 Suggestion for Future Works

Building on the findings and limitations of this study, several directions for future work can improve the predictive modeling of in-hospital mortality in mechanically ventilated CHF patients. Future research should test the models on independent datasets from different healthcare systems. This would help confirm the models' accuracy and reliability across various patient groups, hospitals, and locations. Adding time-series data, like changes in vital signs or lab results over time, could better capture how a patient's condition evolves. Using advanced time-series modeling techniques, such as recurrent neural networks (RNNs) or attention-based models, could reveal patterns that improve predictions and make the models easier to understand.

Although this study used SHAP to explain the model's predictions, more work is needed to create tools that make these explanations clear and useful for healthcare professionals. Collaborating with clinicians during development and deployment would ensure the models are practical and meet real-world needs.

Additionally, future research could explore using other types of data alongside the current dataset. For instance, integrating image or text data, such as medical imaging or clinical notes, could give a more complete view of a patient's health and improve prediction accuracy. By addressing these areas, future work can make predictive models more accurate, practical, and helpful for improving patient care.

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7. APPENDIX

7.1 ICD Codes of 72 types of CHF

ICD Code	ICD Version	Title
39891	9	Rheumatic heart failure (congestive)
40201	9	Malignant hypertensive heart disease with heart failure
40211	9	Benign hypertensive heart disease with heart failure
40291	9	Unspecified hypertensive heart disease with heart failure
40401	9	Hypertensive heart and chronic kidney disease, malignant, with heart failure and with chronic kidney disease stage I through stage IV, or unspecified
40403	9	Hypertensive heart and chronic kidney disease, malignant, with heart failure and with chronic kidney disease stage V or end stage renal disease
40411	9	Hypertensive heart and chronic kidney disease, benign, with heart failure and with chronic kidney disease stage I through stage IV, or unspecified
40413	9	Hypertensive heart and chronic kidney disease, benign, with heart failure and chronic kidney disease stage V or end stage renal disease
40491	9	Hypertensive heart and chronic kidney disease, unspecified, with heart failure and with chronic kidney disease stage I through stage IV, or unspecified
40493	9	Hypertensive heart and chronic kidney disease, unspecified, with heart failure and chronic kidney disease stage V or end stage renal disease
4254	9	Other primary cardiomyopathies
4255	9	Alcoholic cardiomyopathy
4257	9	Nutritional and metabolic cardiomyopathy
4258	9	Cardiomyopathy in other diseases classified elsewhere
4259	9	Secondary cardiomyopathy, unspecified
4280	9	Congestive heart failure, unspecified
4281	9	Left heart failure
42820	9	Systolic heart failure, unspecified

ICD Code	ICD Version	Title
42821	9	Acute systolic heart failure
42822	9	Chronic systolic heart failure
42823	9	Acute on chronic systolic heart failure
42830	9	Diastolic heart failure, unspecified
42831	9	Acute diastolic heart failure
42832	9	Chronic diastolic heart failure
42833	9	Acute on chronic diastolic heart failure
42840	9	Combined systolic and diastolic heart failure, unspecified
42841	9	Acute combined systolic and diastolic heart failure
42842	9	Chronic combined systolic and diastolic heart failure
42843	9	Acute on chronic combined systolic and diastolic heart failure
4289	9	Heart failure, unspecified
I099	10	Rheumatic heart disease, unspecified
I110	10	Hypertensive heart disease with heart failure
I130	10	Hypertensive heart and chronic kidney disease with heart failure and stage 1 through stage 4 chronic kidney disease, or unspecified chronic kidney disease
I132	10	Hypertensive heart and chronic kidney disease with heart failure and with stage 5 chronic kidney disease, or end stage renal disease
I255	10	Ischemic cardiomyopathy
I420	10	Dilated cardiomyopathy
I425	10	Other restrictive cardiomyopathy
I426	10	Alcoholic cardiomyopathy
I427	10	Cardiomyopathy due to drug and external agent
I428	10	Other cardiomyopathies
I429	10	Cardiomyopathy, unspecified
I43	10	Cardiomyopathy in diseases classified elsewhere
I50	10	Heart failure
I501	10	Left ventricular failure, unspecified
I502	10	Systolic (congestive) heart failure
I5020	10	Unspecified systolic (congestive) heart failure
I5021	10	Acute systolic (congestive) heart failure

ICD Code	ICD Version	Title
I5022	10	Chronic systolic (congestive) heart failure
I5023	10	Acute on chronic systolic (congestive) heart failure
I503	10	Diastolic (congestive) heart failure
I5030	10	Unspecified diastolic (congestive) heart failure
I5031	10	Acute diastolic (congestive) heart failure
I5032	10	Chronic diastolic (congestive) heart failure
I5033	10	Acute on chronic diastolic (congestive) heart failure
I504	10	Combined systolic (congestive) and diastolic (congestive) heart failure
I5040	10	Unspecified combined systolic (congestive) and diastolic (congestive) heart failure
I5041	10	Acute combined systolic (congestive) and diastolic (congestive) heart failure
I5042	10	Chronic combined systolic (congestive) and diastolic (congestive) heart failure
I5043	10	Acute on chronic combined systolic (congestive) and diastolic (congestive) heart failure
I508	10	Other heart failure
I5081	10	Right heart failure
I50810	10	Right heart failure, unspecified
I50811	10	Acute right heart failure
I50812	10	Chronic right heart failure
I50813	10	Acute on chronic right heart failure
I50814	10	Right heart failure due to left heart failure
I5082	10	Biventricular heart failure
I5083	10	High output heart failure
I5084	10	End stage heart failure
I5089	10	Other heart failure
I509	10	Heart failure, unspecified
P290	10	Neonatal cardiac failure

Table 7.1: ICD Codes for Congestive Heart Failure